GEK-83783

DRIVE SYSTEM INSTRUCTIONS

POWER SYSTEM STABILIZER
3S7932LA300G1 2

NOTE: Includes torsional filter, monitoring circuit, and speed transducer for a toothed-wheel with a magnetic pickup for a 50 Hz bus-fed exciter.

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GENERAL ELECTRIC

POWER SYSTEM STABILIZER

3S7932LA300G12

For A

50HZ BUS FED EXCITER

(GEK-83783)

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CHAPTER 1

1 -1. INTRODUCTION

The Power System Stabilizer, PSS, is designed to introduce a supplementary control signal into continuously acting voltage regulator equipment. The equipment will supplement the conventional voltage regulator control with a control function derived so that, under deviations of generator speed or load, excitation becomes controlled as a composite function of voltage and frequency. Under certain conditions, the composite function has been shown to be an aid to power system stability.

The PSS receives a signal from a toothed wheel with magnetic pick-up located at this turbine front standard. This signal, which is frequency-modulated, is passed through a speed transducer which then has an output frequency equal to the modulation frequency and an amplitude proportional to the deviation in frequency. This signal is then passed through two lead-lag stages in order to obtain the proper phase shift characteristic, a band reject filter to attenuate signals at torsional frequencies, a washout stage to eliminate the effect of component drift and slow system frequency variations, a gain stage to obtain maximum control action and an output stage which interfaces the PSS with continuously acting voltage regulating equipment.

The Power System Stabilizer includes the functions of input transducer, signal conditioning, limiting output circuitry, attenuation of torsional frequencies and torsional protective circuitry. This circuit trips the PSS whenever excessive amplification of torsional frequencies by PSS is detected, or whenever the monitoring circuit fails. Figure 3 is a block diagram of the system, while Figure 13 shows a typical functional elementary diagram.

The Power System Stabilizer will apply the following transfer function to modify the output of the frequency transducer for insertion into the voltage regulator:

$$G(s) = \frac{K\,(T_1S)\,(1 + T_2S)\,(1 + T_4S)}{(1 + T_1S)\,(1 + T_3S)\,(1 + T_5S)}$$

Adjustments are provided to vary the parameters of the PSS so as to permit compensation for variations in the characteristics of the regulator, excitation system, generator and power system to which it is applied. These adjustments can be made by calibrated dial settings which will be accurate to within 5% of the desired parameter value. The adjustments are capable of modifying the transfer function parameters through the following typical ranges: K = 1 to 100

T1 = 0.1 to 50 seconds
T2 = 0.2 to 1.5 seconds
T3 = 0.02 to 0.1 seconds
T4 = 0.2 to 1.5 seconds
T5 = 0.02 to 0.1 seconds

1-2. RECEIVING AND HANDLING

The Power System Stabilizer is normally supplied mounted in either the regulator or exciter cubicle. The instruction manual for the particular excitation system used should be consulted for receiving and handling instructions. If the PSS is shipped separately (as in the case of a retrofit system), then immediately upon receipt, the equipment should be carefully unpacked to avoid damage. It should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric sales office should be notified promptly.

I-3. STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If equipment is to be stored for more than three months, it should be kept heated to at least 5°C above ambient temperature to prevent condensation.

1-4. ABBREVIATIONS

The following abbreviations are used in this instruction manual:

AC	Alternating Current
amp	Amperes
db	Decibels
DC	Direct Current
FET	Field Effect Transistor
HZ	Hertz
IC	Integrated Circuit
LED	Light Emitting Diode
MA	Milliampere
MV	Millivolt
PP	Peak-to-Peak
PT	Potential Transformer
PSS	Power System Stabilizer
SCR	Silicon Controlled Rectifier
VAC	Alternating Current Voltage
v c o	Voltage-Controlled Oscillator
VDC	Direct Current Voltage
°C	Degrees, Centigrade
RPM	Recolutions per minute
BFE	Bus-Fed exciter

1-5. DESCRIPTION

The Power System Stabilizer consists of:

3S7932LA300G12 Assembly

The Power System Stabilizer is normally mounted in the regulator cubicle and contains the following:

NUMBER	NAME	QTY.
44C372607G01	4800 HZ Speed Transducer	1
44C331857G03	Signal Conditioning Board	2
44C372640G01	Band Reject Filter	1
44C372640G02	Band Pass Filter	1
44C331859G01	Washout and Output Board	1
44C331863G02	Relay Board	1
44C331861G01	Ceiling Sensing Board	1
44C372641G01	PSS Comparator	1

1-6. INSTALLATION

I-6.1. Location and Connections

The Power System Stabilizer is normally furnished as an integral part of the excitation and regulation equipment. Refer to the drawings furnished with the excitation system and the instruction manual for the excitation system for location and connection instructions.

I-6.2. initial Checks

1. Wire check all external connections to the Power System Stabilizer. Be sure that wire size and type are adequate for the expected signal levels. Refer to the connection diagram furnished with the excitation equipment.

CAUTION

DO NOT HI-POT OR MEGGER ANY CIRCUIT, AS THIS MAY RESULT IN DAMAGE TO COMPONENTS. ANY WIRING OR CIRCUITS TO BE HI-POTTED OR MEGGERED MUST FIRST BE DISCONNECTED FROM THE PSS.

2. Ensure that all printed circuit boards are fully inserted.

1-7. OPERATION

CAUTION

MISOPERATION OF THE POWER SYSTEM STABILIZER WHILE CONNECTED TO THE EXCITATION SYSTEM MAY RESULT IN DAMAGE TO THE GENERATOR. DEVIATIONS FROM THE FOLLOWING OPERATIONAL CHECKS AND PROCEDURES SHOULD BE CONSIDERED IN LIGHT OF THIS AND SHOULD BE DISCUSSED WITH THE MANUFACTURER PRIOR TO INITIATION.

1-7.1. Initial Operation

CAUTION

THE POWER SYSTEM STABILIZER MUST BE OUT OF SERVICE OR THE REGULATOR IN THE DC MODE WHEN CONDUCTING ALL TESTS THRU PARAGRAPH 10. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

1-7.1 .1. Magnetic Pick-up Tests. With the turbine generator at rated speed, check the output of the magnetic pickup with an oscilloscope. This can be done by connecting the oscilloscope leads across the input terminals or connecting the leads to "IN TEST" on the **Speed Transducer** board (Loc. 10501). For a 3000 RPM machine and a 96 toothed wheel, this frequency will be 4800 Hz. The signal should be approximately a sine wave (some distortion is mormal) and have a magnitude of at least 5 volts PP (normally 30 to 40 volts PP will be obtained). Amplitude modulation of the signal will exist because of shaft run-out, imperfect toothed wheels, etc. This usually appears as a once per revolution frequency (60 Hz for a 3600 RPM machine). The existence of high frequency spikes usually indicates a shield grounding problem and a check should be made for the existence of more than one ground on the shield. Pull the **Speed Transducer** board.

I-7.1.2. Power Supply Tests. If the PSS does not use the cubicle power supplies, it is necessary to check the built-in power supplies. This does not apply to the Generrex* Excitation System. Check elementary diagrams on other excitation systems.

Turn on +15 VDC power using the toggle switch mounted on the faceplate of the **Power Supply** board.

Check that the power supply voltages are correct. Adjustment of the + 15 volt supply may be made with board mounted rheostats. Refer to Theory of Operation section. The current limit and the crow bar settings are factory set and need no further adjustment.

If, as on the Generrex Excitation System, the cubicle power supplies are used, ensure that the power supply busses are energized and at the proper level. If adjustment is needed, refer to the excitation system instruction book.

CAUTION

ON SOME EXCITATION EQUIPMENT, PULLING THE RELAY BOARD INTRODUCES A STEP INTO THE REGULATOR. THIS CAN RESULT IN DAMAGE TO THE GENERATOR. ENSURE THAT THE REGULATOR IS IN THE DC MODE OF OPERATION BEFORE PULLING THE RELAY BOARD.

1-7.1.3. Relay Board Tests. If the generator is in the AC mode of operation, transfer to DC. Pull the Relay board (LOC 10512). The Relay board is designed to interface with many excitation systems. Board-mounted shorting clips and rheostats are used to adapt the Relay board to the particular excitation system. Refer to the drawings furnished with the equipment for the settings of F1S, F2S, F3S, F4S, F1P and F2P. Check settings and readjust if necessary. Insert **Relay** board. Pull **Washout and Output** board (LOC 10508). Put shorting clip C5S on terminals 1, 2, "not gnd". Insert **Extender** board in LOC 10508.

CAUTION

THE REGULATOR MUST BE IN THE DC MODE OF OPERATION FOR THIS TEST. FAILURE TO PLACE THE REGULATOR IN THE DC MODE MAY RESULT IN DAMAGE TO THE GENERATOR.

Insert a signal from pins 11, 12 to pins 3, 4 of the **Extender** board. This signal **can** be a DC signal or an oscillating signal. Peak value should be less than 10 volts. With the PSS off, as evidenced by the "OFF" LED on the **Relay** board being lighted, no voltage should exist at the test jack labeled "Out to Regulator" mounted on the Relay board. Pressing the "ON" pushbutton should cause the voltage to be present but attenuated for insertion into the appropriate regulator. If the "ON" LED will not stay energized, press the "Reset" pushbuttons mounted on the faceplate of the **Ceiling Sensing** board and the faceplate of the PSS Comparator board. If the "ON" LED will still not stay energized, check that the permissive contact is closed. This is usually 52G line breaker. If the unit is off line, this permissive contact must be jumpered in order to establish that the output relay is working properly. Press the "OFF" pushbutton. The voltage at the test jack should be zero and the "OFF" LED should be lighted. Remove Extender board. Insert **Washout and Output** board (LOC **10508).** On **Ceiling Sensing** board (LOC 10510), put the delay dial on zero, the positive limit dial on 5 and the negative limit dial on 5. Press the reset pushbutton on the **Ceiling Sensing** board. Adjust the gain for 1.0 by turning the dial marked "Gain K" on the **Washout and Output** board to zero. Connect a variable source of DC voltage to "IN1" of the **Washout and Output** board. Put C5S jumper on 1 and 2. Set the voltage to some value less than 5 volts. Press the "ON" pushbutton on the **Relay** board. A voltage should exist at the test jack on the **Relay** board labeled "Out to Regulator". Increase the voltage at "IN1" of the **Washout and Output** board to 6 volts. After approximately 2.5 seconds, the FAR relay should drop out causing the "OFF" LED to be lighted. Try **to** reinsert the PSS by pressing the "ON" pushbutton. The PSS should not latch in the "on" position. Remove the voltage from IN1. Press the reset pushbutton **on** the **Ceiling Sensing** board. Put the PSS in service by pressing the "ON" pushbutton. The PSS should latch in the "on" position and the "ON" LED should be lighted. Remove the PSS from service by pressing the "OFF" pushbutton. The "OFF" LED should be lighted. Customer controls for the PSS are normally mounted in the control room. Check that these controls can be used to insert and remove the PSS. Leave the PSS in the "off" position.

CAUTION

THE FOLLOWING GAIN TEST MUST BE PERFORMED WITH THE PSS IN SERVICE AND THE REGULATOR IN THE AC MODE. ENSURE THAT THE SHORTING CLIPS AND RHEOSTAT SETTINGS ON THE RELAY BOARD ARE CORRECT FOR THIS EQUIPMENT. ENSURE THAT THE CONTINUOUS WAVE FREQUENCY TRANSDUCER IS PULLED. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

The **Relay** board contains the necessary interface circuitry so that a one volt change in PSS output voltage will cause a one percent change in generator terminal voltage. With the PSS off, a small step of voltage (≈ 1 VDC) should be inserted into IN1 of the **Washout and Output** board. The regulator should be in the automatic mode of operation with the generator preferably off-line. The test may be satisfactorily performed on-line if the reactive current compensator (if any) is bypassed or switched **to** zero and if the voltage regulator of any other generator units paralleled at terminal voltage are temporarily turned to DC control. Place the PSS in service. Measure the change in the generator terminal voltage. The gain is calculated by dividing the percent change in generator terminal voltage by the step in voltage applied to IN1. The gain should be (-) 1%/volt $+0.1$. The minus sign, which means that terminal voltage decreases for a positive input voltage, arises because the signal is inserted into IN1, which bypasses the washout stage. F1P on the **Relay** board can be used to adjust the gain if necessary. Remove the voltage at IN1.

An attempt has been made to interface with the voltage regulator in such a way that placing the PSS into and out of service will not cause a bump. On some equipments, F2P on the **Relay** board has been furnished for this purpose. With the voltage regulator in DC, or Test, if a test position is available on the control switch, pull the **Speed Transducer** (LOC 10501). Put the PSS in service using the pushbuttons on the **Relay** board. Note whether the AC regulator output voltmeter or transfer voltmeter changes when the PSS is put in service. Use F2P **on** the **Relay** board to minimize the change in AC regulator output voltage when the PSS is inserted.

Remove the PSS from service.

I-7.1 .4. Speed Transducer Test. To check the

operation of the **Speed transducer(LOC.** 10501) disconnect the leads from the toothed wheel/magnetic pickup at L23, L24. Connect an audio oscillator to L23, L24. Adjust the oscillator for approximately 10 volts peak-to-peak at 4800 Hz.

Insert the **Speed Transducer** board. The output of the **Speed Transducer** board should be **zero** (the dc level should be zero but there will be ≈1 volt of 4800 Hz carrier feedthrough). If the voltage is more than ±2.5 VDC, use the null potentiometer, A1P, to get 0 volts ± 0.2 volts. Increase the frequency of the audio oscillator to 5040 Hz. The transducer output should increase by + 5 volts from the 4800 Hz value (± 0.5 volts). Decrease the audio oscillator frequency to 4560 Hz and the transducer output should change by -5 volts from the 4800 Hz value (± 0.5 volts). The gain should **not** be adjusted unless the oscillator frequency calibration is accurate and the output is off by more than 0.5 volts. Remove the audio oscillator. Reconnect the leads from the toothed wheel/magnetic pickup to terminals L23, L24. With the turbine-generator synchronized, adjust the null potentiometer, A1P, for 0 volts ± 0.2 volts at the **Speed Transducer** output test jack. The **Speed Transducer** power supplies are isolated by 12 volt Zeners. If the null cannot be adjusted to 0 **volts ± 0.2** volts, remove one of the shorting clips on the P1 terminal board mounted on the back of the rack. If the minimum voltage attainable is positive, then remove the shorting clip from Pl-7 to Pl-6. If the minimum voltage is negative, remove the shorting clip from Pl-5 to Pl-6.

CAUTION

THE PSS MUST BE OUT OF SERVICE FOR THE FREQUENCY RESPONSE TESTS. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

1-7.1 .5. Frequency Response of Main Channel. Record pot settings and jumper positions prior to testing so that control can be reset.

Set both Signal Conditioning boards (LOC 10502, 10503) for lead time constants of 0.26 seconds and lag time constants of 0.03 seconds. This can be accomplished by the following adjustments:

TCT2V = 0 turns
TCT2F = 0.9 turns
TCT3 = 1.5 turns
TCT4 = 0.9 turns
TCT5 = 1.5 turns

BIS jumper should be on 1-2 (GRD).

On the **Washout and Output** board (LOC 10508), set the washout time constant for 2 seconds and the gain for 20. This can be accomplished by placing the TCT1 dial on 6.8 and the gain control at 3.8.

Jumpers should be as follows:

C1S 2-3
C2S 2-3 (150K)
C3S l-2 (l-50)
C4S l-2 (l-50)
C5S 2-3 (GRD)

Check the **Band Reject Filter** (LOC 10504) to confirm that the shorting clips on 1TB, 2TB, 3TB and 4TB are on terminals 1, 2. The shorting clip on 5TB should be on terminals 2, 3. Pull the **Speed Transducer** board (LOG 10501). Insert the **Extender** board in place of the **Transducer** board. Connect an audio oscillator to terminal 11, 12 and ground terminal 3 or 4 of the **Extender** board. Measure the frequency response of the PSS main channel from 0.1 Hz to 100 Hz, measuring oscillator output and the output of the **Washout and Output** board. The latter can be measured on the "Out 1" test jack of the **Washout and Output** board. The oscillator output level should be held to those shown in Table 1 to avoid saturation of the amplifiers. Record data and compare gain to values shown in Table 1. Actual gain and nominal gain should not differ by more than 2 to 1.

Input and output signals should be observed with an oscilloscope to be sure that saturation or distortion is not present.

1-7.1.6. Frequency Response of Monitor Channel . Using the same settings as in paragraph 5, make a frequency response check of the monitor, or protective, channel. Set gain pots 11P and 31P on the **PSS Comparator** (LOC 10506) to minimum, fully CCW. The signals of concern are the voltages at 2TP and 3TP of the PSS Comparator board.

Use an oscilloscope to make sure saturation or distortion is not present.

Set the audio oscillator frequency for 15 Hz. Balance the voltages at 2TP and 3TP on the **PSS Comparator** board by increasing the lesser of the two voltages. Use either 11P or 31P. Using Table 2 as a guide, determine the amplitude response of 2TP and 3TP as a function of frequency. The amplitude of the voltage at 2TP and 3TP should not differ by more than 0.25 volts over the frequency range shown in Table 2. The gain at 2TP should be calculated and compared to the nominal gain shown in Table 2. The actual gain and nominal gain should not differ by more than 2 to 1. The nominal gain as a function of frequency is shown in Figure 1 for the main channel. and the protective channel.

1-7.1.7. PSS Comparator Tests. Check out the trip circuit of the **PSS Comparator.** The trip reference voltage at 4TP is five times the voltage difference in 2TP and 3TP at which a trip will occur. Set the trip reference adjustment, 51P, for 2.5 volts at 4TP. With the oscillator at 15 Hz, slowly adjust, in either direction, TCT2F on either **Signal Conditioning** board until the voltage at **5TP** of the **PSS Comparator** begins to rise. The voltage at 2TP and 3TP should differ by 0.4 to 0.6 volts. The trip

Figure 1. Frequency Response of Main Channel

FREQ	HZ	V(OUT) VOLTS (ZERO TO PEAK)	GAIN	NOMONAL GAIN (1)
0.1	0.2			14.1
0.2	0.2			18.2
0.4	0.2			24.8
1.0	0.2			58.9
2.0	0.02			175.8
3.0	0.02			295.1
5.0	0.02			446.7
7.0	0.02			421.7
10	0.02			234.4
12	0.02			102.3
15	0.2			10.0
17	0.2			6.8
20	0.2			3.2
25	0.2			1 8
30	0.2			1.1
35	0.2			1.3
40	0.2			1.9
45	0.2			1.9
50	0.2			2.1
55	0.2			2.1
60	0.2			4.7
65	0.2			6.3
70	0.2			5.7
80	0.2			6.5
90	0.2			7.8
100	0.2			8.9

T, = 2 SEC.; T_2 = T_4 = 0.26 SEC.; T_3 = T_5 = 0.03 SEC.; GAIN = 20

(1) Gain is measured from the output of the Saeed Transducer to the outut of the Washout and **Output** board. Use an oscilloscope to ascertain that nothing is saturating.

light should begin to flash as 5TP goes positive. Return TCT2F to its original position and reset the trip.

Final settings of the **two Signal Conditioning** boards should be identical. The final setting of the **PSS Comparator** trip reference, 51P, should be coordinated with the infinite shaft life torque levels of the machine. Refer to the manufacturer for this setting. Remove the audio oscillator.

I-7.1.8. Ceiling Sensing Tests. Check out the **Ceiling Sensing** board. Pull the **Ceiling Sensing** board and verify that the shorting clip on E1S is between 1 and 2. Reinsert the board. Set both the positive limit, E2P, and negative limit, E3P, for 500, (5.0 turns). Set the time delay on the **Ceiling Sensing** board for minimum, (0). Insert a variable DC voltage into IN1 of the **Washout and Output** board. Place the regulator in the manual mode of operation.

CAUTION

THE REGULATOR MUST BE IN THE DC MODE OF OPERATION FOR THIS TEST. FAILURE. TO DO THIS CAN RESULT IN DAMAGE TO THE GENERATOR.

Place the PSS in service by pressing the "ON" pushbutton located on the Relay board. Slowly increase the voltage until the PSS trips out of service as evidenced by the "OFF" LED being lighted. The voltage at OUT1 on the **Washout and Output** board should be 5 volts + 0.25 volts. Remove the voltage. Pressing the "ON" pushbutton on the **Relay** board should have no effect. Press the reset pushbutton on the **Ceiling Sensing** board. Pressing the "ON" pushbutton on the Relay board should now cause the PSS to be placed in service. Reverse the voltage into IN1 of the **Washout and Output** board. The same procedure can be followed to verify that this part of the circuit is working. Adjust time delay E1P on the **Ceiling Sensing** board for 300, (3 turns). Remove the voltage at IN1 of the **Washout and Output** board and press the reset pushbutton on the **Ceiling Sensing** board. Place the PSS in service, **with the** regulator in manual. Put a voltage into IN1 of sufficient magnitude to cause a trip. The PSS should trip to "OFF" approximately 10 seconds after insertion of the signal. Reset the **Ceiling Sensing** board by pressing the reset pushbutton on the faceplate. Place the positive and negative limit dials on 1000, (10 turns), and the time delay dial on zero. Final settings can only be made after on-line test and/or system studies.

I-7.1.9. Washout and Output Limit Tests. On **Washout and Output** board, set negative limit dial and positive limit dial on 500, (5 turns). Insert a variable DC voltage into IN1. Monitor the output voltage at OUT1 using an oscilloscope or voltmeter. Increase the DC voltage slowly and note the voltage level at which the output fails to follow the input voltage (output voltage is limited). This voltage should be 4.75 volts + 0.5 volts. Reverse the voltage and perform the same test. The output voltage should be of the opposite polarity but same magnitude, within + 0.5 volts. Remove the voltage and **Extender** board. Reinsert the **Speed Transducer** board. Ensure that the PSS is off. Place the regulator in the AC mode of operation. Set the positive and negative limit dials on 1000, (10 turns) and the time delay dial on zero. Final settings can only be made after on-line tests and/or system studies.

I-7.1.10. Maximum Gain Noise Tests. The adjustment range provided by the PSS can result in extremely high gain over a range of frequencies. For instance, with K set at 100, T2 and T4 set at maximum and T3 and T5 set at minimum, the maximum gain will be 120 db., which is 1,000,000:1. Although this gain will

probably never be used in an actual system, a good check of PSS noise level can be obtained using this gain.

Ensure that the PSS is not in service. On the **Signal Conditioning** board (LOC 1102), set TCT2F and TCT4 for 1000, (10 turns) and TCT3 and TCT5 for zero. Pull the **Washout and Output** board (LOC 10508). Put the shorting clip on C5S on terminals 2, 3, "GRD". Put the shorting clip on C3S on terminals 2,3. Insert **Washout and Output** board. Turn gain adjustment to 1000, (10 turns). Remove the **Speed Transducer** (LOC 10501) and insert the extender in its place. Short from pin 15 to pin **3** of the **Extender** board. Using an oscilloscope with a very slow sweep speed, such as 1 sec/cm, determine the peak-to-peak output voltage, measured at OUT1 of the **Washout and Output** board. This voltage should be less than 3 volts peak-to-peak. Remove **Extender.** Insert the **Speed Transducer** board in LOC 10501. **Remove Washout and Output** board. Place C5S shorting clip on terminals 1, 2, "not grd". Place C3S shorting clip on terminals 1, 2. Return gain adjustment dial to zero which is unity gain. Insert **Washout and Output** board. Final settings of time constants and gain can only be made after on-line tests and/or system studies.

I-7.2. Initial Settings

The PSS has been designed to interface with various excitation systems. In order to permit compensation for widely different systems, adjustments are made with calibrated dials. On the **Washout and Output** board, set the gain dial to zero and C3S on l-2. Ensure that the **Relay** board has the correct rheostat and switch settings.

Settings for a particular installation are best obtained by analytical studies and/or on-line tests. It is the responsibility of the customer to perform the analytical studies and on-line tests necessary to determine proper settings. General Electric Company can provide assistance in performing analytical studies and system tests. Contact your local sales office. A paper, "Concepts of Synchronous Machine Stability as Affected by Excitation Control", by F. P. deMello and C. Concordia may be used as a reference (68-TP-129-PWR). Other publications which may be used as references are, "Test Procedures for Power System Stabilizers," published by the Western Systems Coordinating Council, and an IEEE publication, "Guide for the Identification, Testing and Evaluation of the Dynamic Performance of Excitation Control Systems," paragraph 6.03 (P421, 1977).

I-7.3. On-Line Tests

Power system stabilizers are employed to provide an additional input to the regulator to improve power system dynamic performance. They have been employed to solve two separate but related problems; local machine-system oscillations in the l-2 Hz range and inter-area oscillations of 0.5 Hz or less. The fundamental function of the signal conditioning network of the PSS is to compensate for the phase lags of the system being controlled. For best contribution to system damping, the gain of the PSS should be maximized under the most sensitive conditions, usually at or near full load. The desired stabilizing signal has as its reference the base system speed which is slowly changing all the time. The washout function serves to minimize the effect of overall system speed changes on machine terminal

Table 2. Frequency Response at Comparator

FREQ. HERTZ	V(IN) PEAK MILLIVOLTS	2TP - MILLIVOLTS	3TP +MILLIVOLTS	DIFF |2TP| - |3TP|	GAIN V(2TP)/V(IN)	NOMINAL GAIN
1	3.0					0.3
2	1.0					1.8
4	0.2					8.4
8	0.2					31 7
10	0.2					25.4
12	0.2					25.2
15	0.2					24.5
18	0.2					22.9
20	0.2					20 4
25	0.2					14.5
30	0.2					8.1
40	0.2					2.4
45	2.0					0.8

NOTE: See instructions for gains and time constants used for nominal gain figures.

2TP-3TP difference should not exceed 0.25 volts at any frequency.

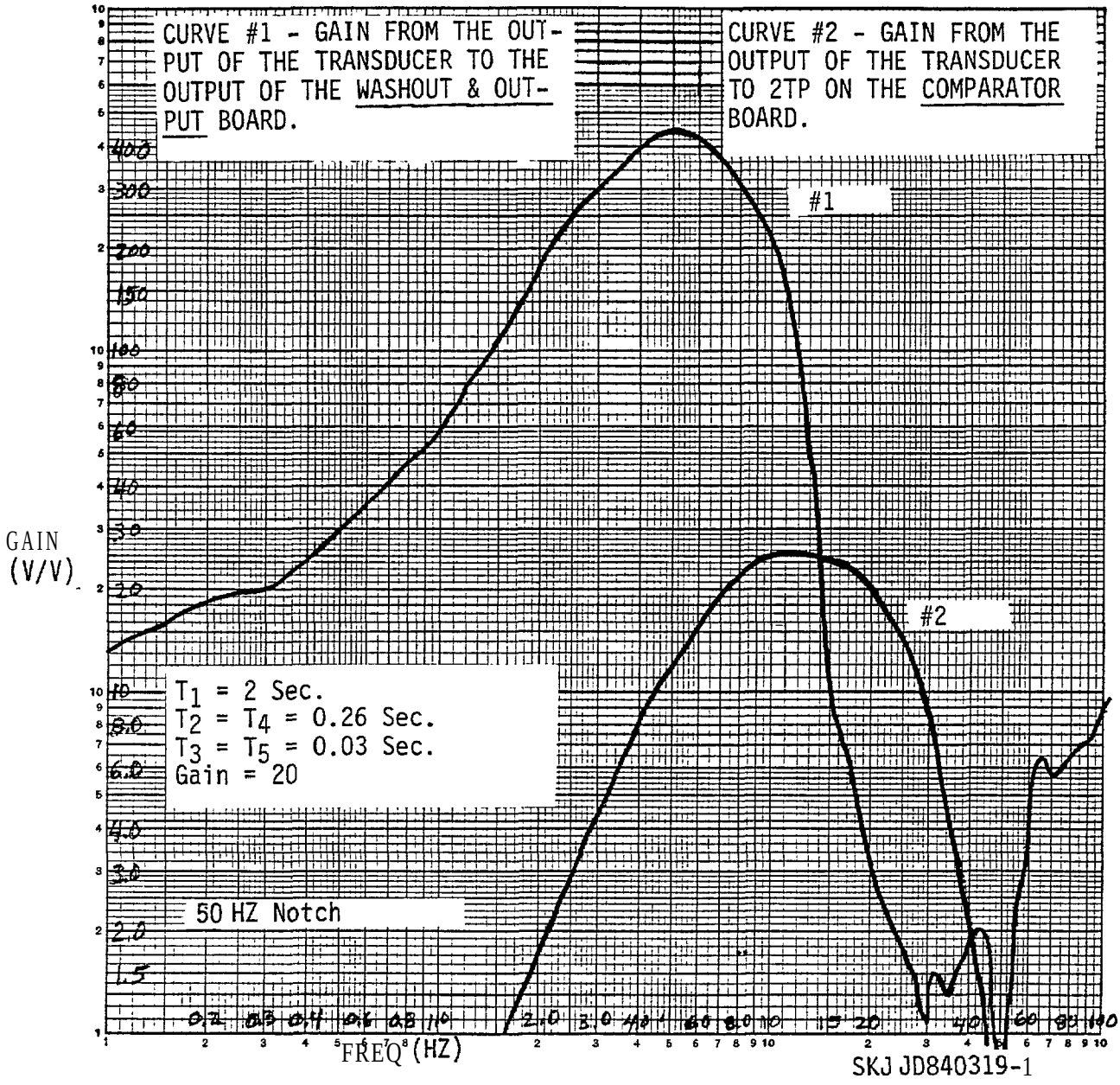


Figure 1. PSS GAIN VS FREQUENCY

voltage.

CAUTION

THE PSS HAS A WIDE RANGE OF ADJUSTMENT OF LEAD AND LAG TIME CONSTANTS. DO NOT ADJUST FOR A DIFFERENCE IN LEAD-LAG SETTINGS OF MORE THAN TEN TO ONE. LARGER DIFFERENCES IN SETTINGS CAN CAUSE EXCESSIVE AMPLIFICATION OF TORSIONAL SIGNALS AND RESULT IN DAMAGE TO THE TURBINE-GENERATOR.

The PSS has adjustable limits on the output signal so that the PSS will not completely override voltage regulator action. The limits are independently adjustable but should be set identically in order to prevent nonsymmetrical limiting which might bias the voltage regulator. Since the ceiling sensing function occurs after the adjustable limit stage, the output limits should be set at a lower voltage than the ceiling sensing voltage. Refer to Figure 2.

In order to facilitate on-line frequency response testing of the excitation system, IN1 is provided on the **Washout and Output** board. With the shorting clip on C5S on terminals 1, 2, a drive signal can be inserted into IN1 which will produce a calibrated terminal voltage change through the regulator. For normal operation, the shorting clip on C5S should be on terminals 2, 3, "GRD" and the shorting clip on C1S should be on terminals 2, 3. Figure 2 is provided so that dial settings can be chosen to realize the desired time constants and limit settings.

I-8. PRINCIPLES OF OPERATION

I-8.1. General

The power system stabilizer is used to provide an additional input to the voltage regulator which, under certain system conditions, can improve power system dynamic stability. A signal is obtained from a toothed wheel/magnetic pickup mounted at the turbine front standard. This signal is frequency modulated with the carrier frequency being 4800 Hz (for a 3000 RPM machine and 96 toothed wheel) and the modulation frequency being equal to the speed changes of the turbine shaft. This frequency modulated signal is passed through a **Speed Transducer** to produce an output signal whose magnitude is proportional to the magnitude of the deviation in speed and whose frequency is equal to the speed deviation frequency. This demodulated signal is then passed through two stages of lead-lag compensation to provide phase shift compensation for the system, a twin-tee notch filter tuned to 50 Hz, a band reject filter to attenuate torsional frequencies, a washout stage to remove drift and slow system frequency changes and an adjustable gain stage to maximize the effect of the PSS. Limits are provided on the output stage so that during large system disturbances, the PSS will not interfere with voltage regulator action. A relay is provided for placing the PSS in service, and also removal from service. Interfacing is provided to properly scale the PSS output signal so that 1 volt from the output stage produces a 1% change in terminal voltage. Protective circuits are provided to detect a failure which causes the output to go to ceiling, **(Ceiling Sensing** board) and a failure which causes excessive amplification of torsional frequencies, (an extra **Signal Conditioning** board, **Bandpass Filter** and **PSS Comparator** board). Ten-turn rheostats with calibrated dials are furnished for all customer adjustments. Figure 3 is an overall block diagram of the PSS.

I-8.2. Power Supply System(Figure 4)

3S7932LA300G12 uses the power supplies from GENERREX. Figure 4 is included and is the normal power supply for the PSS.

The rack mounted PSS utilizes the power supply of the cubicle in which it is mounted. Refer to the instruction manual for the voltage regulator cubicle for a description of the power supply.

The panel mounted PSS includes the ± 15 VDC and 24 VDC power supplies. See Figure 4. 115 VAC, 50 Hz power should be connected to L21 and L22. S1T and S2T transform the input voltage and provide isolation. The voltage is rectified by a single phase full-wave bridge, filtered to remove the AC component, limited by zener diode action and then fed to series pass regulators. Fuses R1F and R2F, SCRs R1CR and R2CR, along with associated sensing and firing circuits form a "crow-bar" circuit which will protect sensitive semiconductor components from overvoltage by blowing a fuse and thus removing the excess voltage. IC voltage regulators G1A and G2A are used to control the output voltage. G2P and G4P, mounted on the printed circuit board, provide adjustment of the output voltage. G1R, G1P, G2R, and G3R provide a current limit feature on the + 15 volt regulator. A similar circuit is found on the -15 volt regulator. The current limit is of the foldback type, i.e., the maximum output current is reduced as the load on the regulator is increased above the full load value. This ensures that critical components in the regulator, such as series pass transistors, will not be damaged by a short circuit. The current limit can be adjusted using G1P and G5P. The positive limit is factory set at 1 amp and the negative limit at 750 MA. These are not the short circuit current values. Integrated circuits J1A and J2A provide voltage sensing for the "crow-bar" circuit. J1P and J2P are factory adjusted to cause the "crow-bar" circuit to operate at a supply voltage of 17 volts. S1T and S2T have taps which will permit variations in the input voltage from the nominal 115 VAC value. Refer to the diagram for the correct tap for a given input voltage. In order to reduce the noise level to as **low** a value as possible, the ± 15 volt power

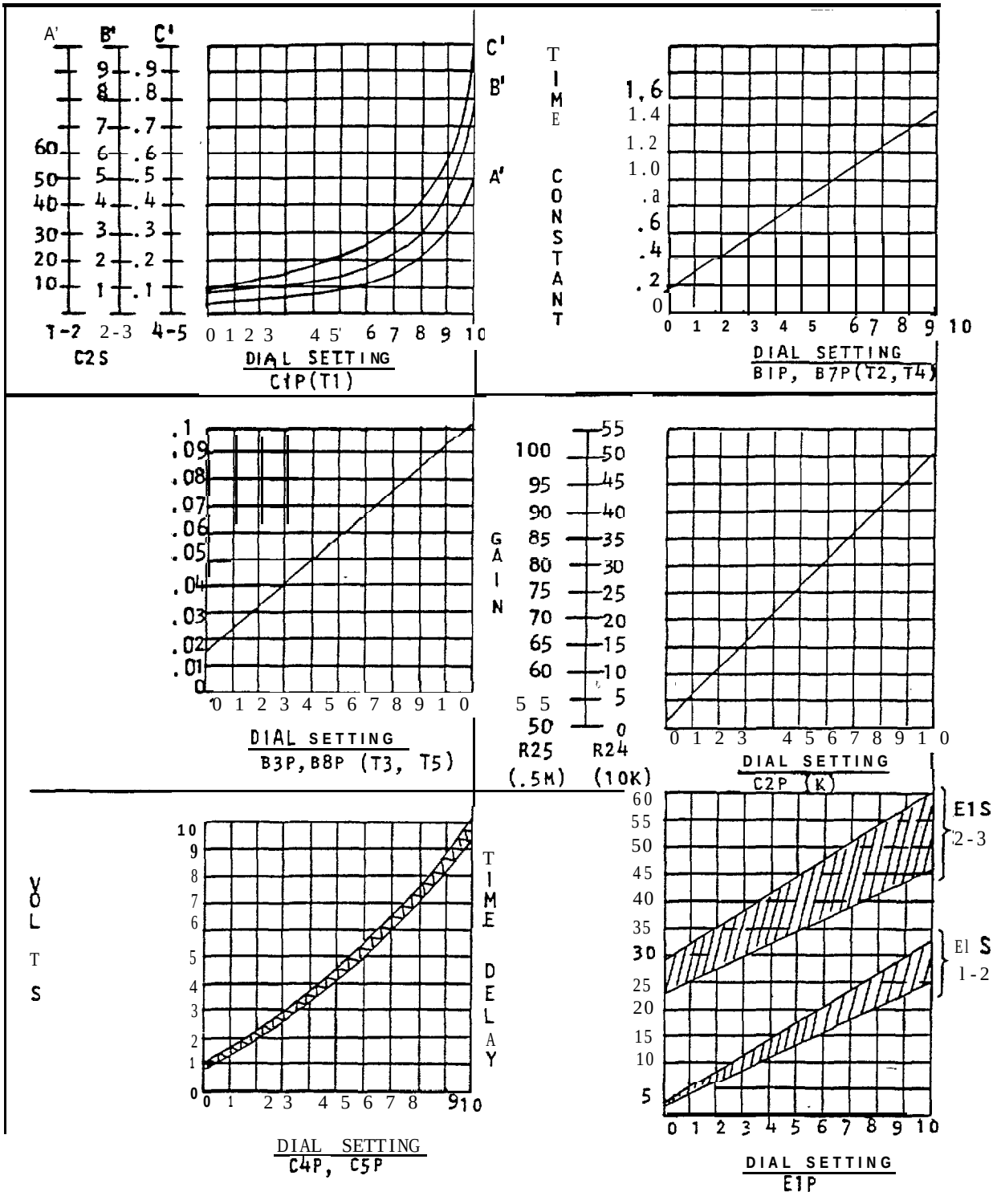


Figure 2. Parameter Value vs Dial Setting

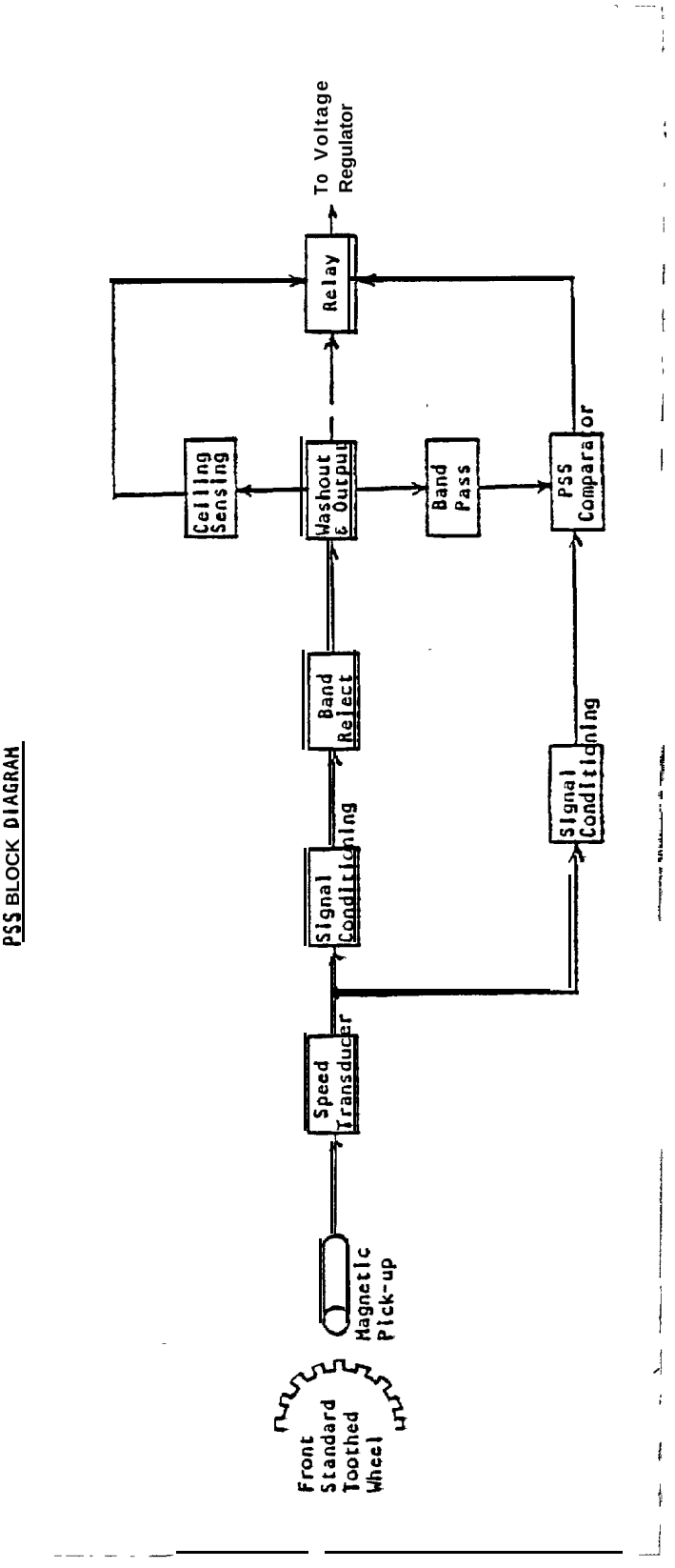


Figure 3. PSS Block Diagram

supply is very tightly regulated and heavily filtered. The regulation is 0.1% and the ripple on the power supply should be ≤5 MV P-P. The 24 volt supply does not require tight regulation since it is used to drive relays and energize lights. S3T is used to isolate the 24 volt supply from the ± 15 volt supply and to transform the input voltage to a lower voltage. This voltage is rectified by a full-wave bridge and then filtered. Regulation is provided by zener diode S1Z. The 24 volt power supply is ungrounded while the ± 15 volt supplies are grounded through a parallel resistor capacitor combination. The high impedance ground on the ± 15 volt supplies allows the PSS to be used with both grounded and ungrounded systems without ground loop problems.

I-8.3. Speed Transducer Board(Figure 5)

The input signal is transformer coupled through 1T. The 1Q is used as a zero crossing detector and with 2Q forms a high gain stage so that whenever a zero crossing is detected, the voltage at 4TP will increase abruptly to 15 volts. This sudden transition in voltage is capacitively coupled through 1C to a monostable multivibrator composed of 3Q, 4Q and associated circuitry. A pulse is generated at 6TP that has an off-time determined by 2C and 8R. During the time that the input signal is negative, 1D protects 1Q and the monostable multivibrator is not triggered. 6TP stays at -5 volts. When the input voltage goes positive by approximately 1 diode drop (VBE), a pulse is generated which triggers the monostable multivibrator and the voltage at 6TP goes to approximately -15 volts. This pulse lasts for a period of time determined by 2C and 8R. The average value of the signal at 6TP is thus determined by the ratio of the on time to off time (on is -5 volts; off is -15 volts). The off time is fixed by 2C and 8R, therefore, the average voltage at 6TP will vary with input frequency. The signal at 6TP is fed through a low pass filter to summing amplifier 1A. A signal is taken from 1P to null the output of 1A at rated frequency so that deviations in frequency are detected. The 2P and 17R are used as feedback for 1A. The 2P is adjusted to give a 1 volt change for a 1% change in input frequency. Provisions have been made so that the feedback time constant can be selected for 280 Hz or 5 Hz. It should be set for 280 Hz on units with a band reject fiter by disconnecting A10C from the circuit by means of the jumper.

I-8.4. Signal Conditioning Board (Figure 6)

The **Signal Conditioning** board consists of two lead-lag stages and a twin-tee notch filter. In addition to the signal from the **Speed Transducer,** an extra input with a separate lead time constant adjustment is provided for a voltage input, IN2. B1P provides adjustment of the first stage lead time constant which is proportional to B1C, B1R and the setting of B1P. B3P provides adjustment of the lag time constant which is proportional to B9R, B4C and the setting of B3P. The DC gain of the stage is B9R/B1R, which is 1.0. B3R, B3C and the equivalent resistance of B3P cause extraneous lead and lag time constants which have been minimized but may be significant when performing calculations at torsional frequencies. The second lead-lag stage is identical to the first except for the addition of an output voltage limit circuit, formed by B1Z, B2Z, B5D and B6D and a drive circuit formed by B1Q, B21R, B22R, B3D and B4D, to increase the output current capability of B2A. A twin-tee notch filter is placed between the first and second lead-lag stages to remove 50 Hz. The twin-tee is factory adjusted to produce greater than 50 db attenuation at the notch frequency. The notch frequency is given by:

$$F = \frac{1}{2\ \pi\ (B11R + B4P)\ (B16C)}$$

I-8.5. Band Reject (Biquad) Filter (Figure 7)

The Band Reject, (Biquad) Filter is a five-stage filter used to attenuate signals at the torsional resonant frequencies of the turbine generator. It provides unity gain outside the rejection band and approximately 40 db attenuation in the rejection band. Figure 8 is a calculated attenuation versus frequency curve. In each stage, amplifiers 11A, 12A and 13A, with associated components, comprise a resonant circuit. The output of 12A is summed with the input signal in 14A to obtain an output signal. Signals not within the pass band of the resonant circuit are passed with no attenuation. Signals that fall within the pass band of the resonant circuit are passed and then subtracted from the original signal to give a reject characteristic.

The transfer function for each stage is described by the following equation:

$$\frac{e_o}{e_i} = \frac{1 + \frac{\alpha_N S}{\omega_o} + \left(\frac{S}{\omega_o}\right)^2}{1 + \frac{\alpha_D S}{\omega_o} + \left(\frac{S^2}{\omega_o}\right)}$$

Where α_N, α_D, and ω_o are determined by the value of the resistors and capacitors used. See Figure 7A.

I-8.6. Band Pass, (Biquad) Filter (Figure 7A)

The **Band Pass, (Biquad) Filter** is used to provide an input for the **PSS Comparator** board so that a check can be made on the **Signal Conditioning** board, **Band Reject, (Biquad) Filter** board, and the washout stage of the **Washout and Output** board relative to amplification of torsional frequencies. The pass filter has an attenuation versus frequency characteristic which is the inverse of the reject filter. By changing two resistors in each stage, the reject filter is changed to a pass filter. By cascading a reject and pass filter, unity gain at all frequencies is obtained. The transfer function for each stage is described by the same equation as for the **Band**

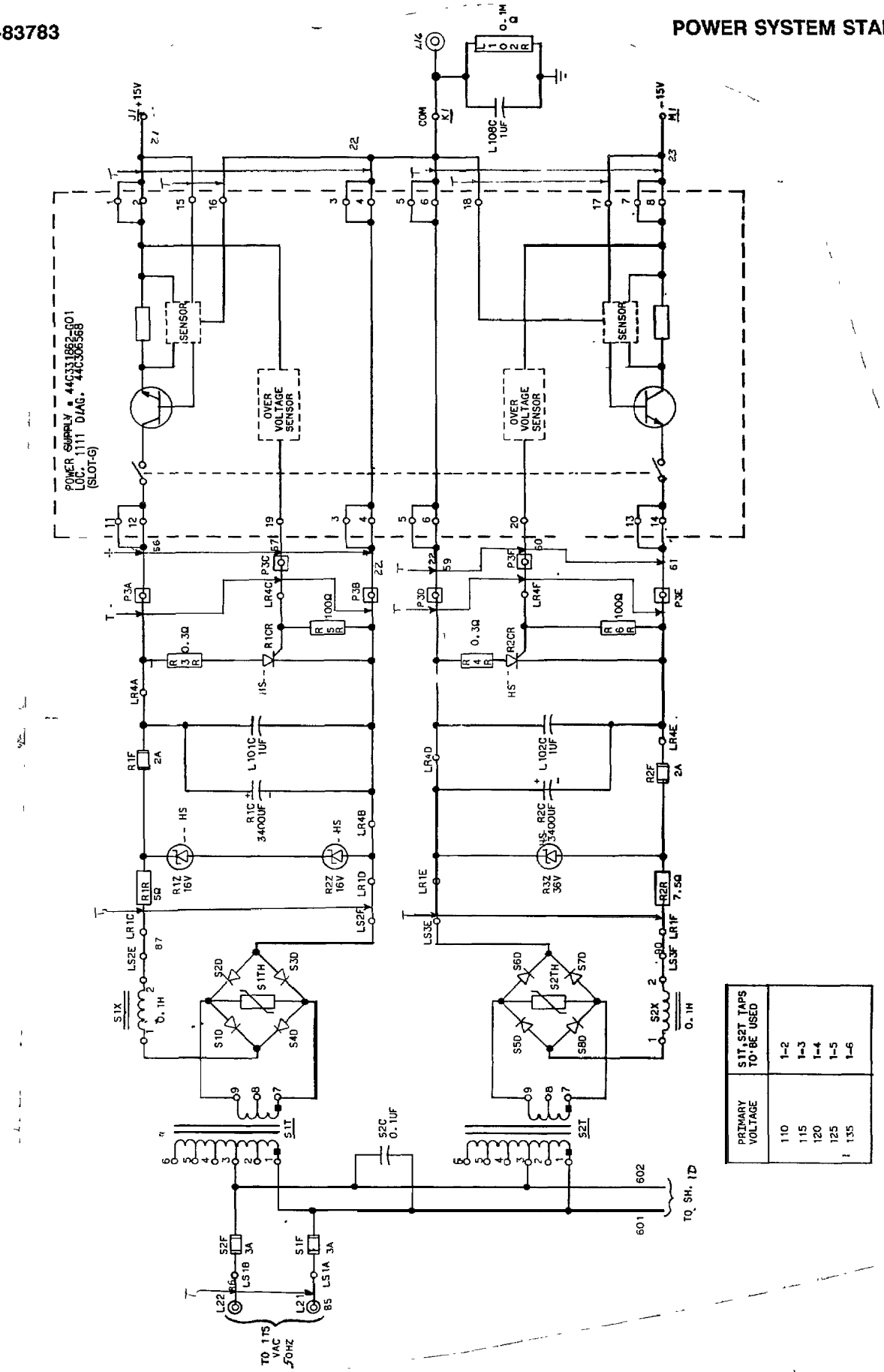


Figure 4. Power Supplies - Panel Mounted

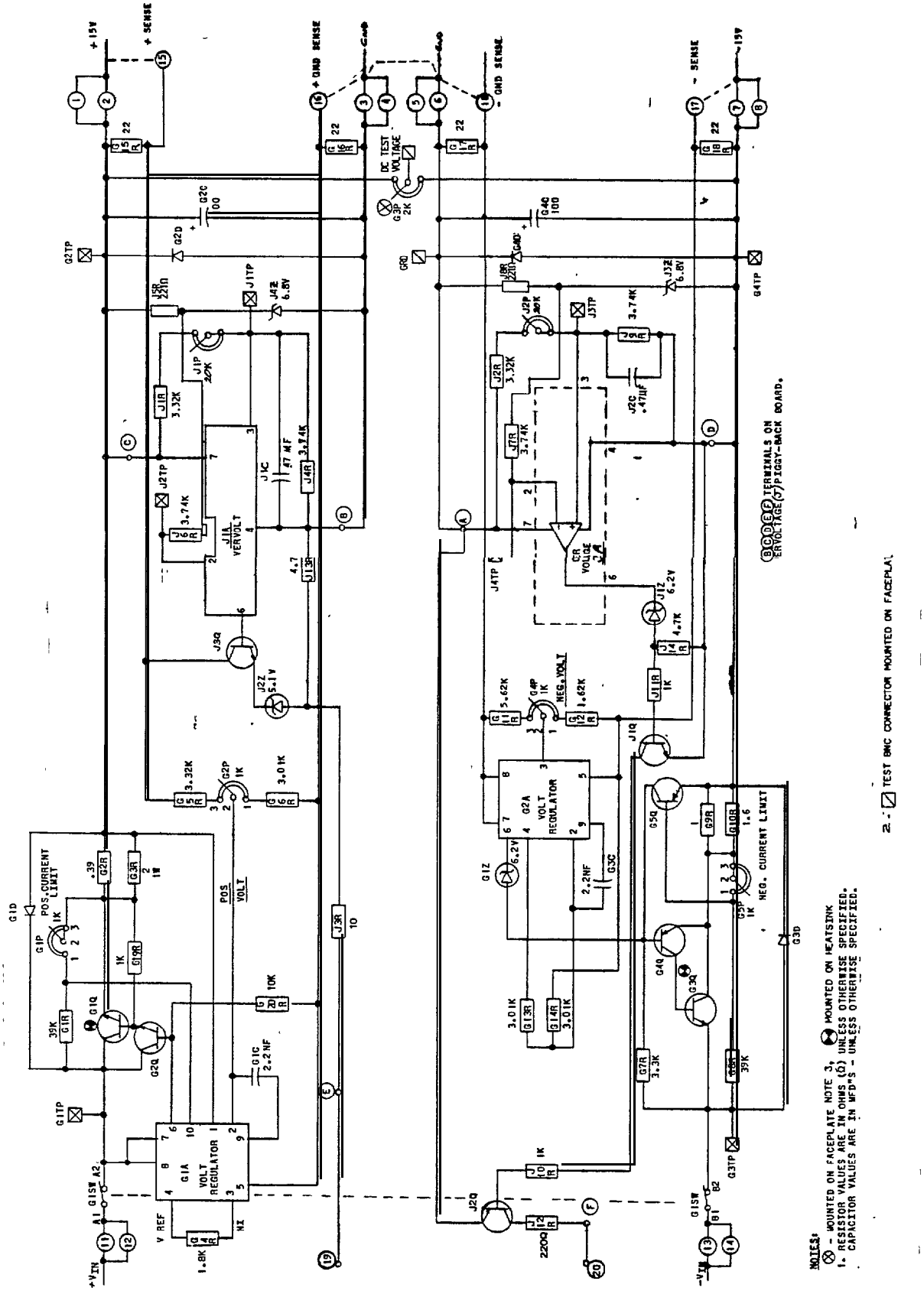


Figure 4A. Sheet 2 ▪ Power Supplies ▪ Panel Mounted

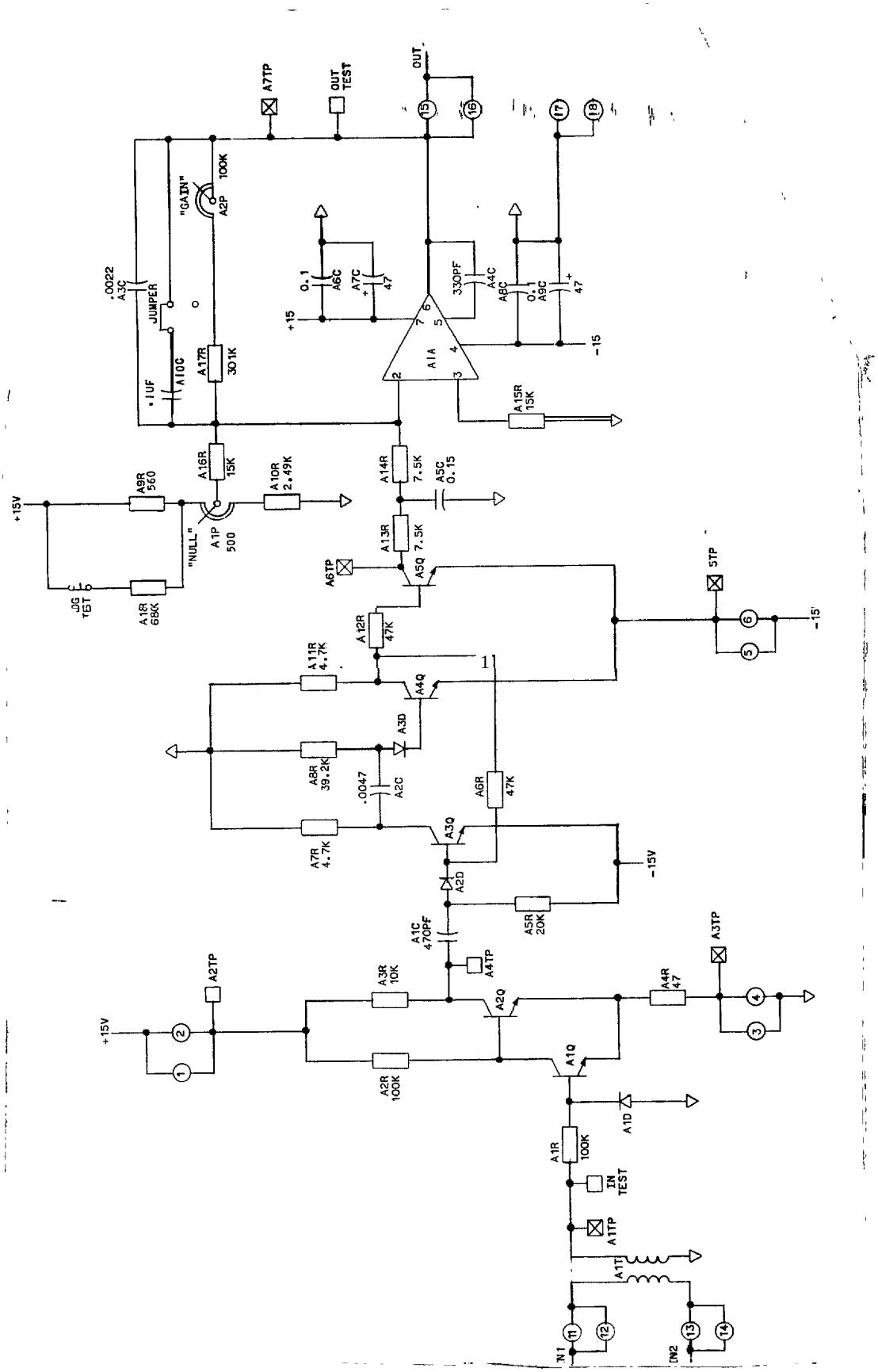


Figure 5. Speed Transducer

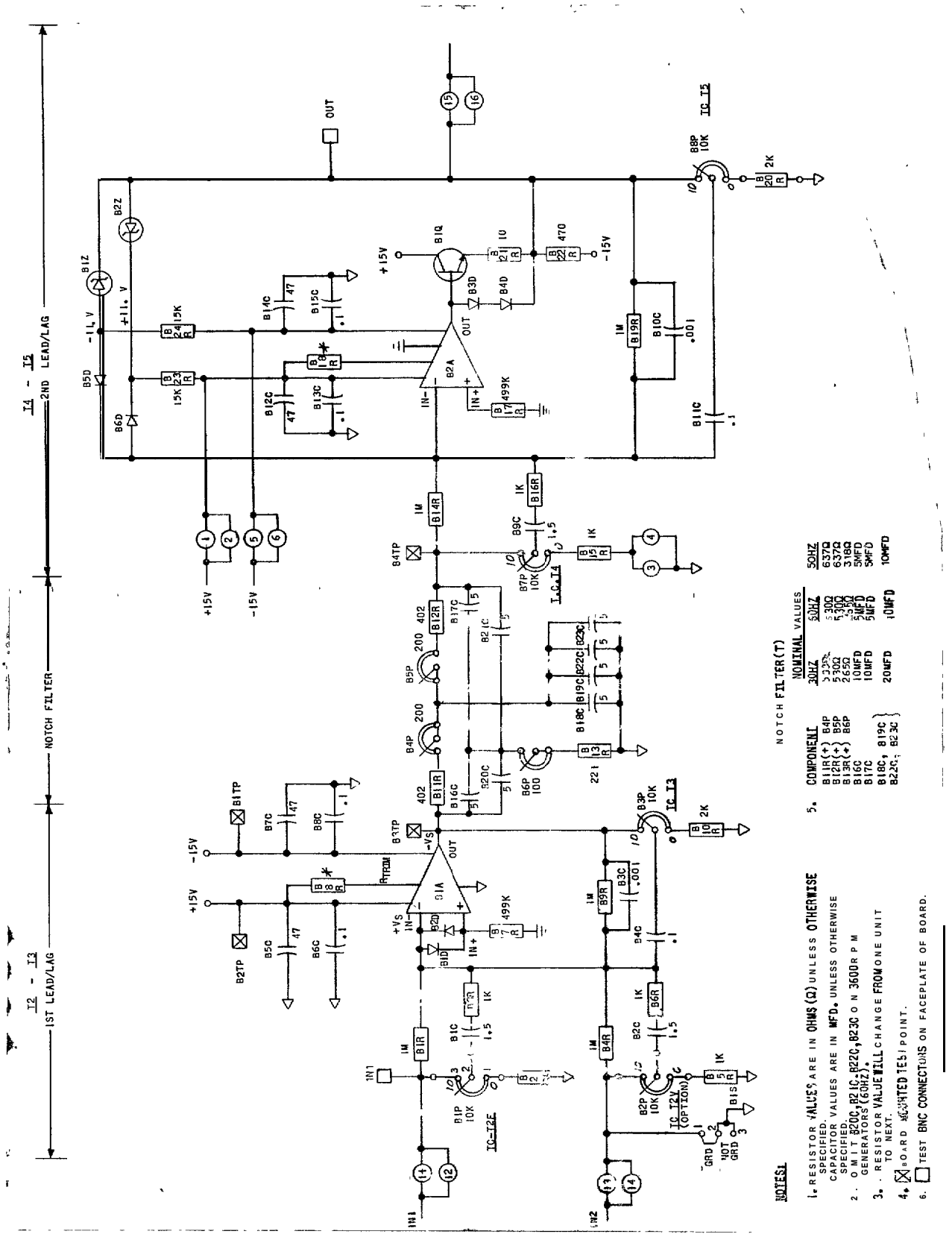


Figure 6. Signal Conditioning

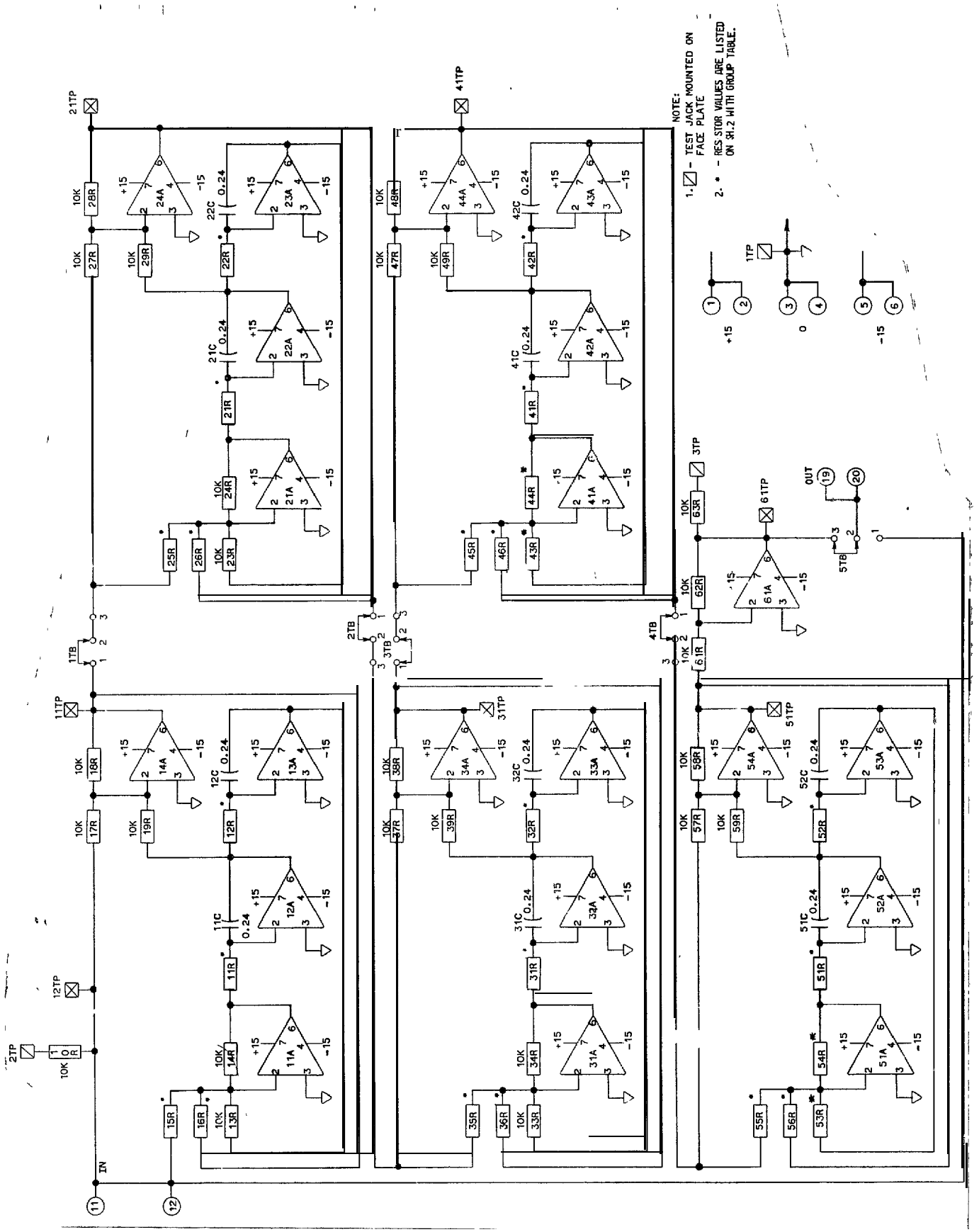


Figure 7. Sheet 1 • Band Reject/Pass Filter

TYPE	B.R.	B.P.	B.R.	B.P.	B.R.	B.P.
11R	43.6K	43.6K	94.7K	94.7K	55.04K	55.04K
12R	43.6K	43.6K	94.7K	94.7K	55.04K	55.04K
15R	83.3K	8.33K	100 OK	10.0K	125.0K	16 87K
16R	8.33K	83.3K	10.0K	100.0K	16 67K	125.0K
21R	35.0K	35.0K	82.8K	82.8K	51.41 K	51.41 K
22R	35.0K	35.0K	82.8K	82.8K	51.41K	51.41K
25R	83.3K	8.33K	50.0K	16.7K	125 0K	12 50K
26R	8.33K	83.3K	16.7K	50.0K	12.50K	125.0K
31R	28.1K	28.1K	73.6K	73.6K	48.05K	48.05K
32R	28.1 K	28.1K	73.6K	73.6K	48.05K	48 05K
35R	83.3K	8.33K	100.0K	10.0K	125.0K	16.67K
36R	8.33K	83.3K	1 0.0K	100.0K	16.67K	125.0K
41R	22.5K	22.5K	44.2K	44.2K	21.59K	21 59K
42R	22.5K	22.5K	44.2K	44.2K	31.09K	31.09K
45R	83.3K	8.33K	50.0K	8.33K	100.0K	8.333K
46R	8.33K	83.3K	8.33K	50.0K	8.333K	100.0K
51R	18.0K	18.0K	31.6K	31.6K	19.18K	19.18K
52R	18.0K	18.0K	31.6K	31.6K	27 62K	27.62K
55R	83.3K	8.33K	33.3K	8.33K	100.0K	8 333K
56R	8.33K	83.3K	8.33K	33.3K	8.333K	100.0K
W01	9 5 6	95.6	44.0	44.0	75.7	75 7
W02	119.0	119.0	50.3	50.3	81.05	81.05
W03	148.3	148.3	56.6	56.6	86.71	86.71
w o 4	185.2	185.2	94.3	94.3	160.8	1608
W05	231.5	231.5	131.9	131.9	181	181
∝D1	1 20	0.12	1.oo	0.10	0.6	0.08
∝D2	1 20	0.12	0.60	0.20	0.8	0.08
∝D3	1 20	0.12	1.00	0.10	0.6	0.08
∝D4	1.20	0.12	1.20	0.20	1.2	0.1
∝D5	1.20	0.12	1.20	0.30	1.2	0.1
∝N1	0.12	1.20	0.10	1.00	0.08	0.6
∝N2	0.12	1.20	0 2 0	0.60	0 08	0.8
∝N3	0.12	1.20	0.10	1.00	0.08	0.6
∝N4	0.12	1.20	0.20	1.20	0 1	1.2
∝N5	0.12	1.20	0.30	1.20	0.1	1.2
∝N1 ∝$\overline{D1}$	1/10	10	1/10	10	2/15	1512
∝N2 ∝$\overline{D2}$	1/10	10	1/3	3	1/10	10
∝N3 ∝$\overline{D3}$	1/10	10	1/10	1 0	2/15	15/2
∝N4 ∝$\overline{D4}$	1/10	10	1/6	6	1/12	12
∝N5 ∝$\overline{D5}$	1/10	10	1/4	4	1/12	1 2 '
43R	10K	10K	10K	10K	8.333K	8.333K
44R	10K	10K	10K	10K	8.333K	8.333K
53R	10K	10K	10K	10K	8.333K	8.333K
54R	10K	10K	10K	10K	8.333K	8 333K

Figure 7A. Sheet 2 - Band Reject/Pass Filter

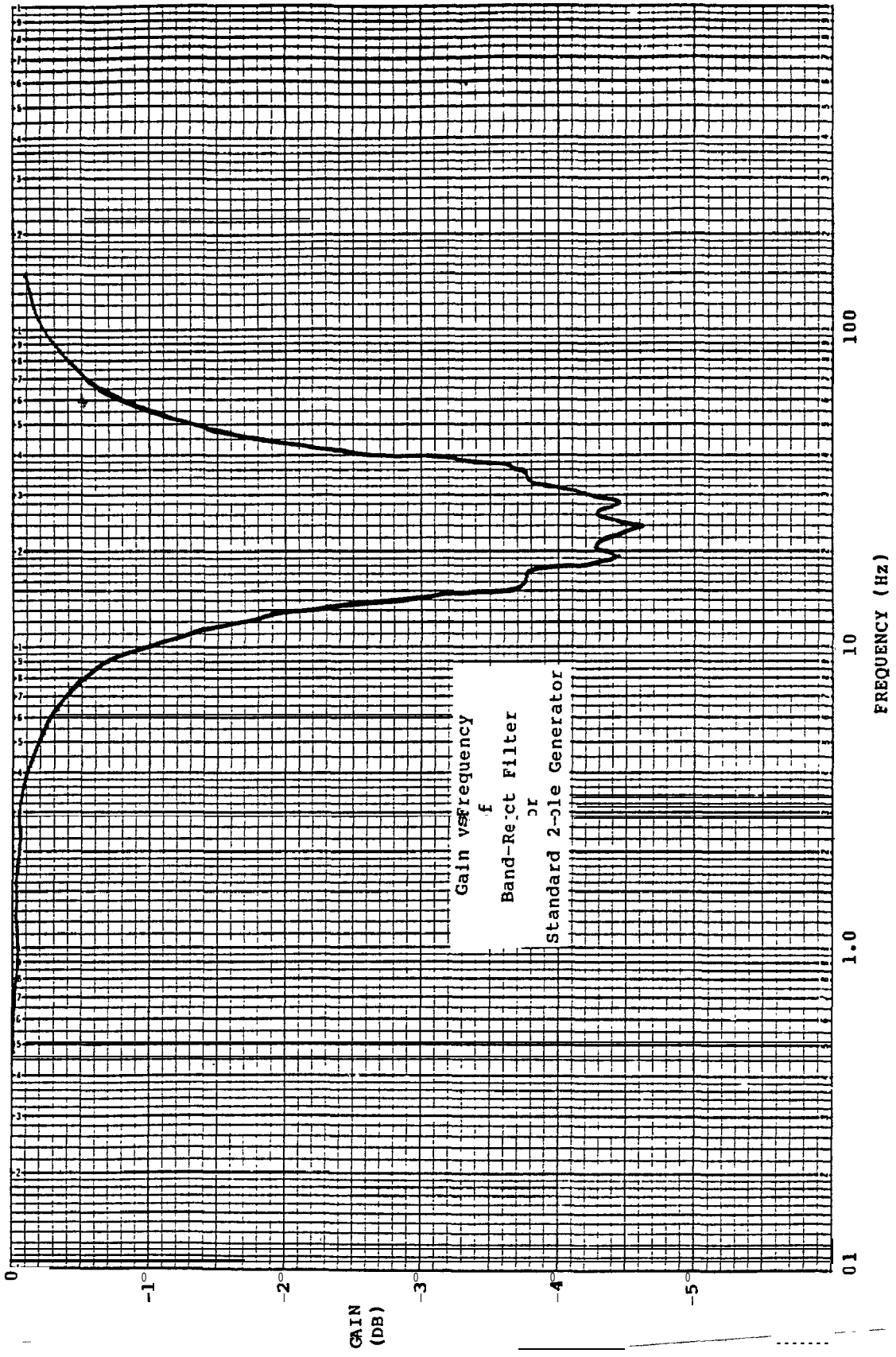


Figure 8. Gain vs Frequency • Band Reject Filter for Standard Z-Pole Generator

Reject, (Biquad) Filter. The values of α_N & α_D are different. See Figure 7A.

I-8.7. PSS Comparator Board (Figure 9)

In order to ensure that the **Band Reject, (Biquad) Filter, Signal Conditioning** board and washout stage are operating correctly, a parallel signal path is provided and the **PSS Comparator** used to detect a failure.

Amplifier 11A is AC-coupled with 12C determining the upper comer frequency. Gain is increased by CW rotation of 11P. Amplifiers 12A and 13A, with associated components, comprise an absolute value circuit whose output is negative. Gain is scaled such that the DC output of 13A equals the peak value of the AC output of 11A. Transistor 11Q with 21R and 22R limit the output from 13A to -10 volts. Amplifiers 31, 32 and 33A operate in the same manner with the exception that the output of 33A is positive and is limited to + 10 volts. The outputs of 13A and 33A supply a net negative current to 52A proportional to the difference in the outputs. If this current exceeds the value supplied from the reference, 52A is driven positive to turn on photocoupler 51Q. In the trip circuit, 51Q and 72Q are connected to form a flip-flop with 51Q normally off. A trip signal turns on 51Q, turns off 72Q, turns on 73Q and 74Q and turns off 75Q, dropping the external relay connected between 15/16 and 9/10. 76Q is turned off. The gates of 71G are connected to form an oscillator which alternately turns 77Q on and off. The indicating lamp, 1LED, flashes for a trip. The trip condition must be manually reset by means of 71PB.

I-8.8. Washout and Output Board (Figure 10)

The **Washout and Output** board contains a washout function, a gain function and a power amplification function. Operational amplifier C1A is a chopper stabilized device for low offset. C1A is basically a derivative circuit formed by C1C and the feedback resistor C2R, C3R or C4R. C2C is added **to** make the derivative circuit only effective for low frequencies and C1P is used to provide adjustment of the effective feedback resistance and thus control the time constant. Coarse adjustment of the time constant is available through C2S. C9D, C10D, Z1C, C31R and C32R form a clamp to limit the output voltage of the washout stage. The output of operational amplifier C1A is used as one of the inputs to the PSS comparator.

An extra input is available at IN1 which bypasses the washout stage. This allows insertion of a signal into the voltage regulator for calibration of the PSS or for frequency response testing of the system. C5S is used to ground IN1 during normal use. Changing C5S shorting clip to terminals 1, 2 allows IN1 to be used for driving the regulator. C2A is used as **an** adjustable gain stage with the gain being determined by C3S and C2P. C3S is used to select a range of gain from 0 to 50 or 50 to 100. An adjustable limit is provided on the output of C2A by C1Q, C2Q, C15R, C16R, C1D, C2D, C4P, C5P, C13R and C14R. The output of C2A is used as the input signal **for Ceiling Sensing** board through pins 13, 14. C3Q and C4Q are junction FETs used as a static switch. In normal operation, C1S should have the shorting clip between terminals 2 and 3, marked "OUT". C3Q will be conducting and C4Q will be non-conducting. C3A is connected to a push-pull transistor stage, (C7Q, C8Q, C9Q and C10Q) in order to increase the output current capability to +100 MA. C3A and the push-pull stage are connected as a unity gain voltage follower.

I-8.9. Relay Board (Figure 11)

The **Relay** board is used to control insertion of the PSS into the regulator, provide the proper interfacing with different voltage regulators and provide interfacing with customer equipment. Meter Ml is mounted on the faceplate and can **be** used to ensure that the PSS output is zero before insertion into the voltage regulator. "OFF" and "ON" pushbuttons are mounted on the faceplate and can be used to insert or remove the PSS from service. LEDs mounted on the faceplate indicate the status of the PSS. A line breaker contact, 52G, should be used as a permissive contact so that the PSS is inoperative off-line. The pushbuttons have momentary contacts. F93 is sealed in by FAR and F93 contacts. F93A is an auxiliary relay used to increase the number of contacts of F93. FAR is controlled by the **Ceiling Sensing** board. The **PSS-Comparator** output static switch is placed in series with the permissive contact, 52G, and is used **to** remove the PSS from service if excessive torsional amplification is detected. F1R, F2R and F1P are used to properly scale the PSS signal for insertion into the regulator. F2P is used to dummy load the regulator so that a bump in excitation will not occur when the PSS is placed in service. Refer to the diagram for the proper settings of F1P and F2P and switches F1S, F2S, F3S and F4S.

1-8.10. Ceiling Sensing Board (Figure 12)

The **Ceiling Sensing** board is used to detect a failure in the PSS which causes the output to be greater than a specified value for an adjustable length of time. Once the condition exists, the PSS cannot be restored to service until a reset pushbutton is depressed on the **Ceiling Sensing** board.

E1A and E2A operate as comparators with E2P and E3P used to adjust the level at which a failure is detected. If the input voltage is less than the positive limit, E2P, and more positive than the negative limit, E3P, then the voltage at E6TP is at a minimum. The voltage at E5TP is passed through an inverter in E1L and then a nand operation is performed on the signal from E6TP and the inverted signal from E5TP. A nand gate has its output in the high state unless all inputs are in the high state.

In this case, both inputs are in the high state so its

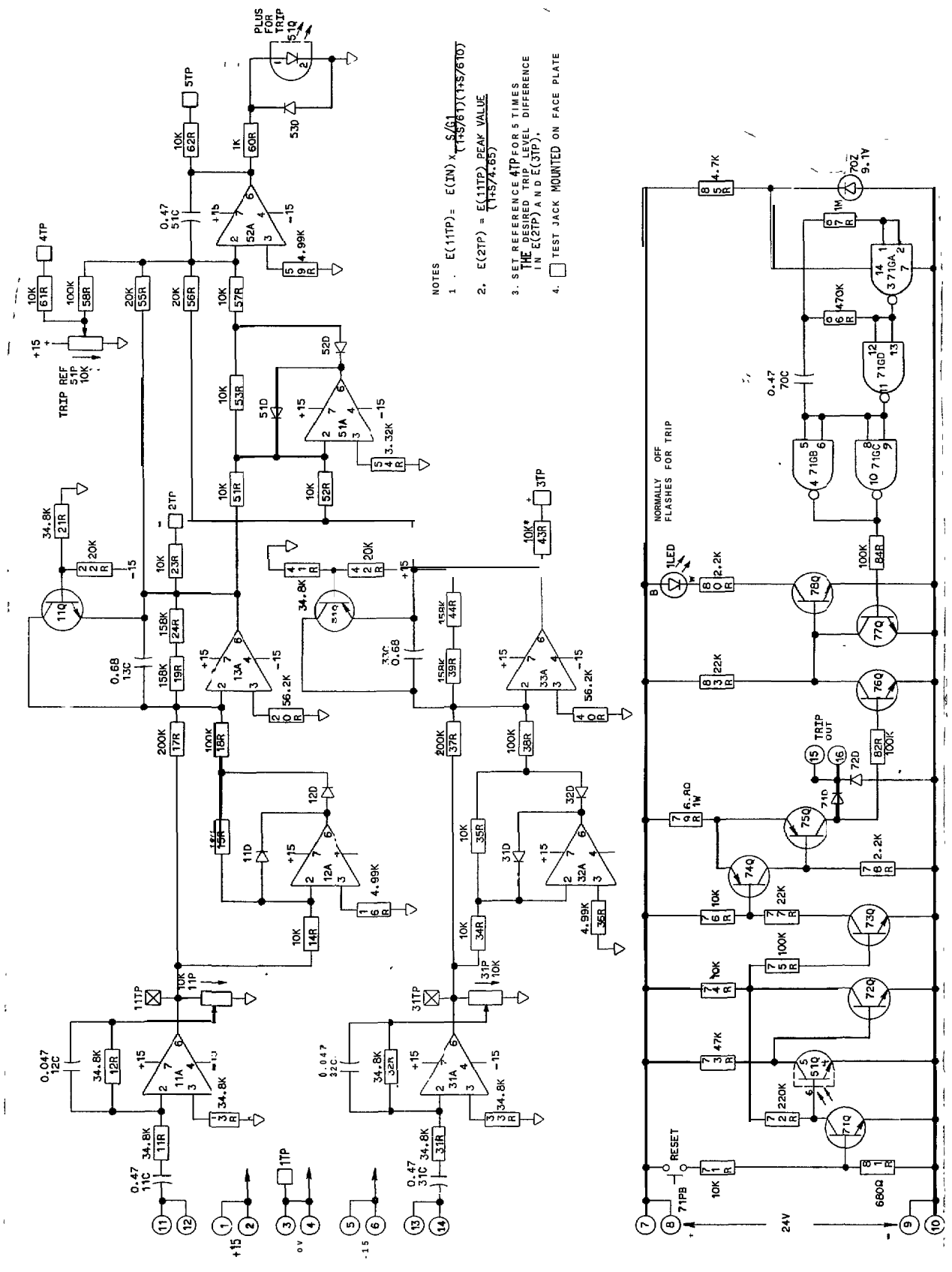


Figure 9. PSS Comparator

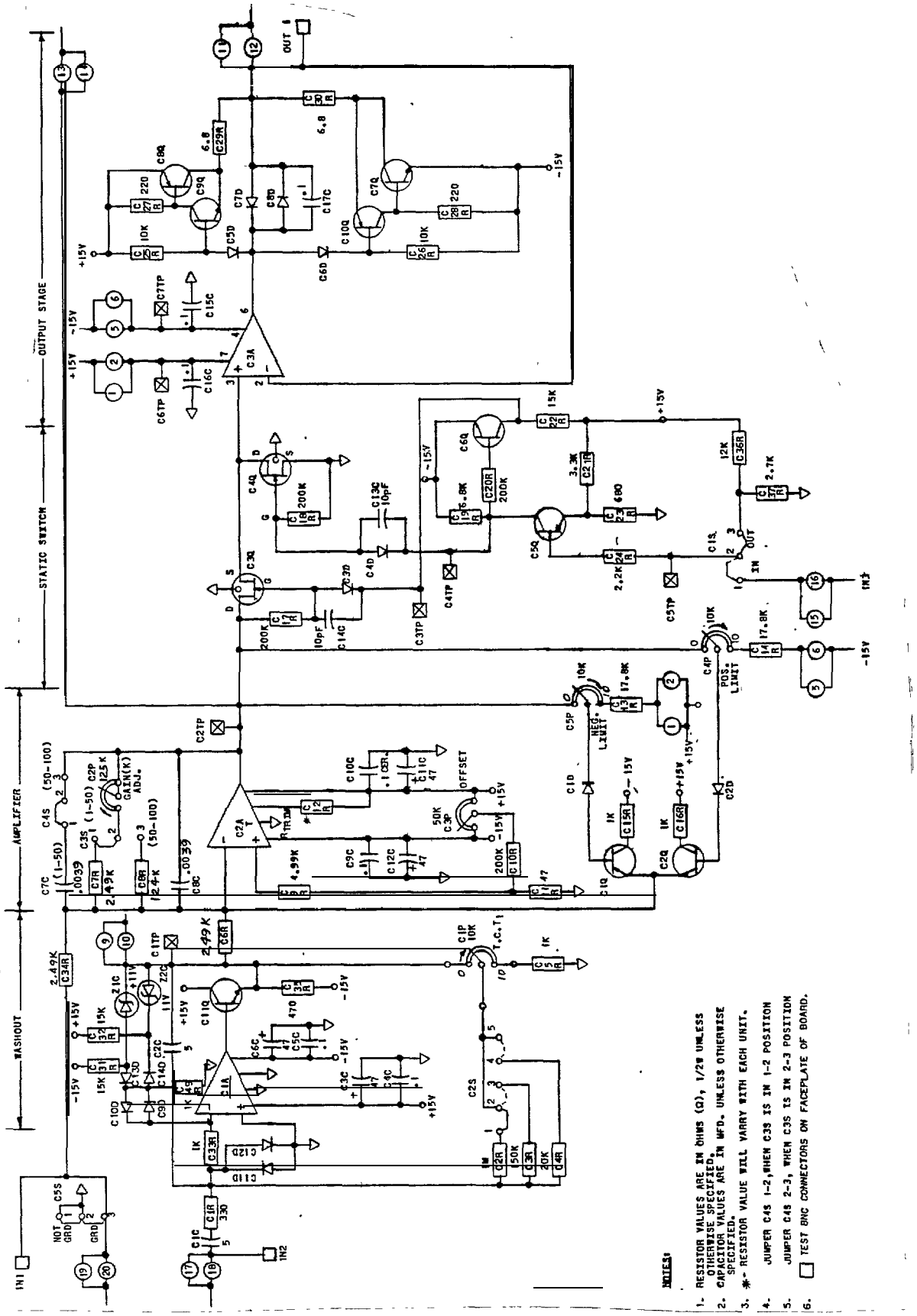


Figure 10. Washout & Output

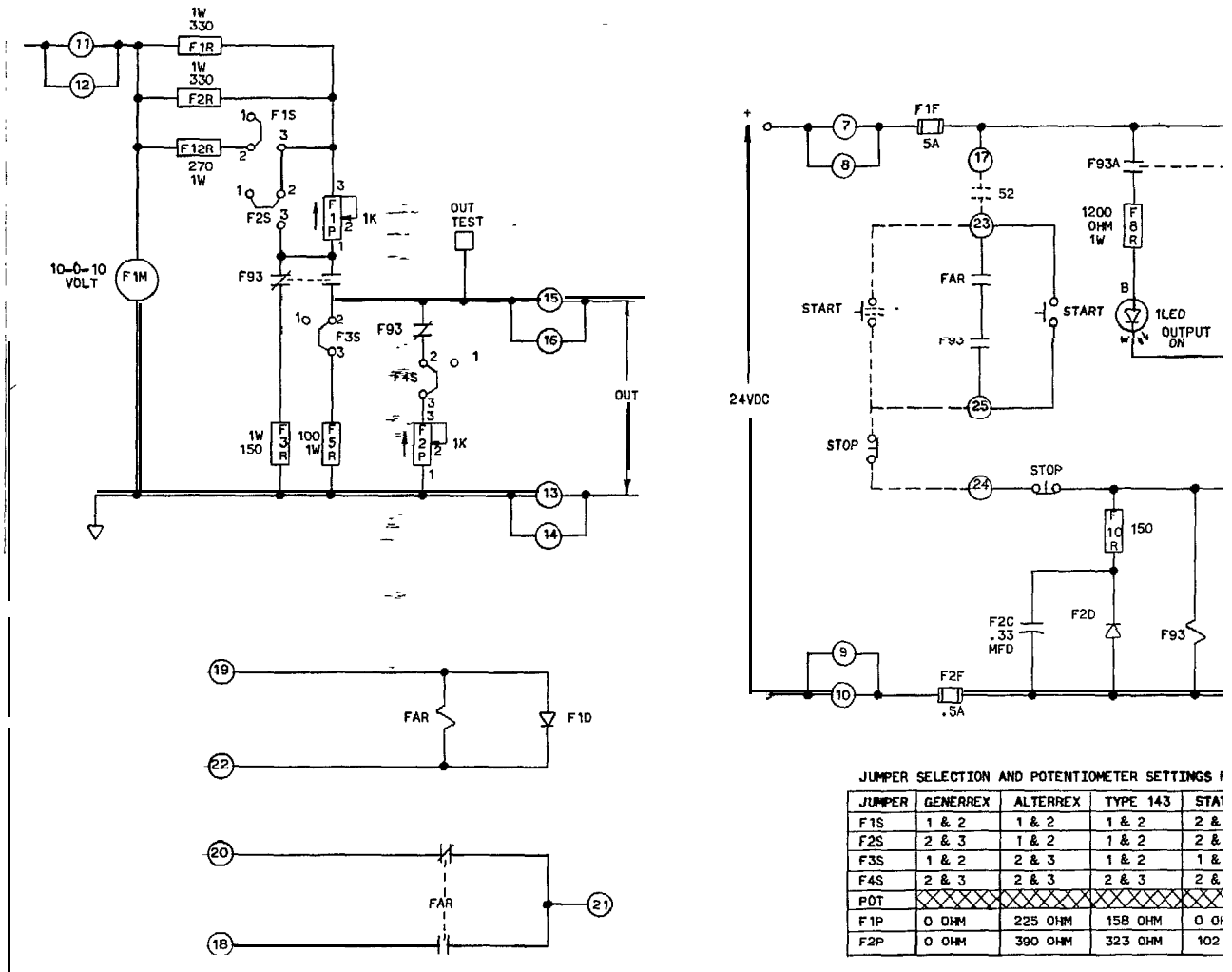


Figure 11. Relay Board

NOTE: RESISTANCE IN OHMS UNLESS OTHERWISE NOTED.

JUMPER SELECTION AND POTENTIOMETER SETTINGS

JUMPER	GENERREX	ALTERREX	TYPE 143	STA
F1S	1 & 2	1 & 2	1 & 2	2 &
F2S	2 & 3	1 & 2	1 & 2	2 &
F3S	1 & 2	2 & 3	1 & 2	1 &
F4S	2 & 3	2 & 3	2 & 3	2 &
POT				
F1P	0 OHM	225 OHM	158 OHM	0 O
F2P	0 OHM	390 OHM	323 OHM	102

output is in the low state. The output of this nand gate is passed through two inverters, resulting in no change in state, and is also used to trigger a one-shot, E3L. With a ceiling condition not present, the one-shot trigger input will stay in the low state and the complemented output stays in the high state. The inputs to the first nand gate of E2L are a low and a high, which means the output will be in the high state. Two of the nand gates are used in a cross-coupled fashion to allow latching whenever a ceiling condition is detected. E8C is provided to ensure that the cross-coupled gates will go to the proper state upon being energized. With no ceiling condition present, the inverter output of E2L will be in the low state, turning on transistors E2Q and E3Q, and energizing relay FAR on the **Relay** board.

If a ceiling condition exists, whether positive or negative, the second nand gate of E1L will have its input states changed to one low and one high. The output of this nand gate will go to the high state which triggers the one-shot, E3L, and changes the input on the first nand gate of E2L to a high state. Nothing happens, however, because the output of the one-shot, E3L, is in the low state and remains there for a period of time determined by E7C, E1P and E8R or E9R, depending on the setting of E1S. After the timing interval, the output of one-shot E3L goes to the high state and the output of the first nand gate of E2L goes to the low state. The output of the cross-coupled gates of E2L goes to the low state and will remain low even if the ceiling condition is cleared. The inverter output of E2L changes to the high state, which turns off E2Q and E3Q and drops out relay FAR on the **Relay** board, giving an alarm and tripping the PSS. To clear the alarm and place the PSS back in service, it is necessary to press reset pushbutton E1PB. This changes the input to one of the cross-coupled gates from a high to a low state which forces the output of the inverter to be in a low state. The cross-coupled gates latch in this state, the relay FAR is picked up and the alarm is cleared.

I-9. MAINTENANCE

The PSS is static, having no moving parts, except for relays and instruments. Therefore, little maintenance should **be** required. If excessive vibration is present, all screw connections should be checked periodically. The **Speed Transducer** null adjustment should be checked periodically and readjusted if necessary. The torsional filter and protective circuit should be checked semi-annually. Refer to the INITIAL OPERATION Section.

I-10. TROUBLESHOOTING

1. If problems are experienced at start-up, carefully check all wiring.
2. If no response is obtained while making the frequency response test:
   a. Check that the power switch on the **Power Supply** board is on, if a **Power Supply** board is furnished.
   b. Check that all boards are fully inserted.
   c. Check for proper jumper connections on filter boards and **Washout and Output** board.
   d. Check each board individually for a response.
3. If an incorrect signal is obtained in terminal voltage, check the switch and rheostat settings on the **Relay** board.
4. If the PSS will not stay in service:
   a. Reset **Ceiling Sensing.**
   b. Check that permissive contact is closed.
   c. **Reset PSS-Comparator**
5. If the **Speed Transducer** output stays at ceiling: Ensure that generator speed is at rated.
6. Readjust null. If a board is found to be defective, replace it and return the defective board for repair and return.

I-11. RENEWAL PARTS

When ordering renewal parts, the following information should be given:

1. Catalog number stamped on the part, with complete description including use and location. Board components should not be replaced. Order the entire board.
2. Complete nameplate data appearing on the assembly that included the component.
3. If possible, data on original order on which equipment was first supplied, including all numerical references.

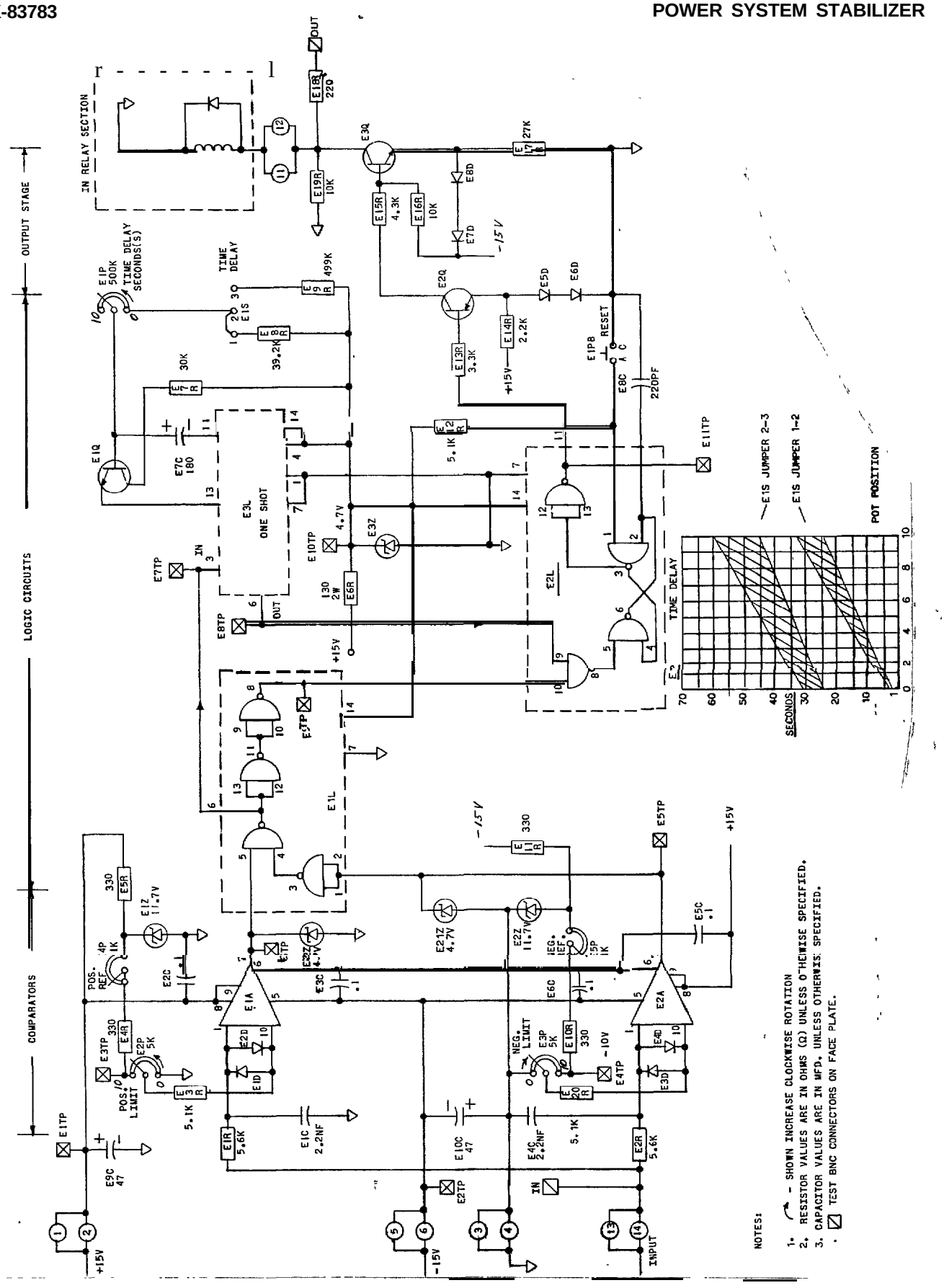


Figure 12. Ceiling Sensing

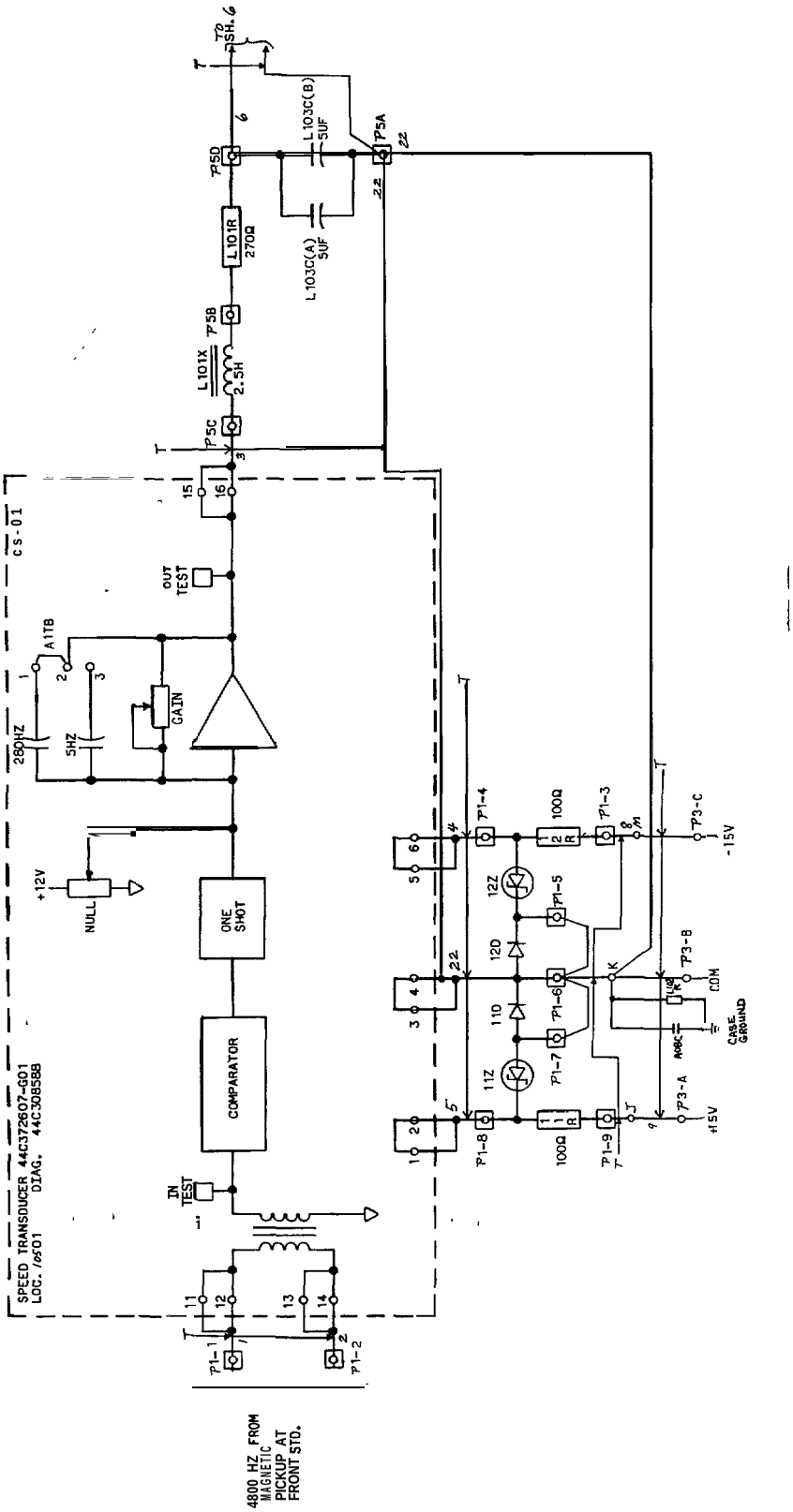


Figure 13. Sheet 5 - Elementary Diagram

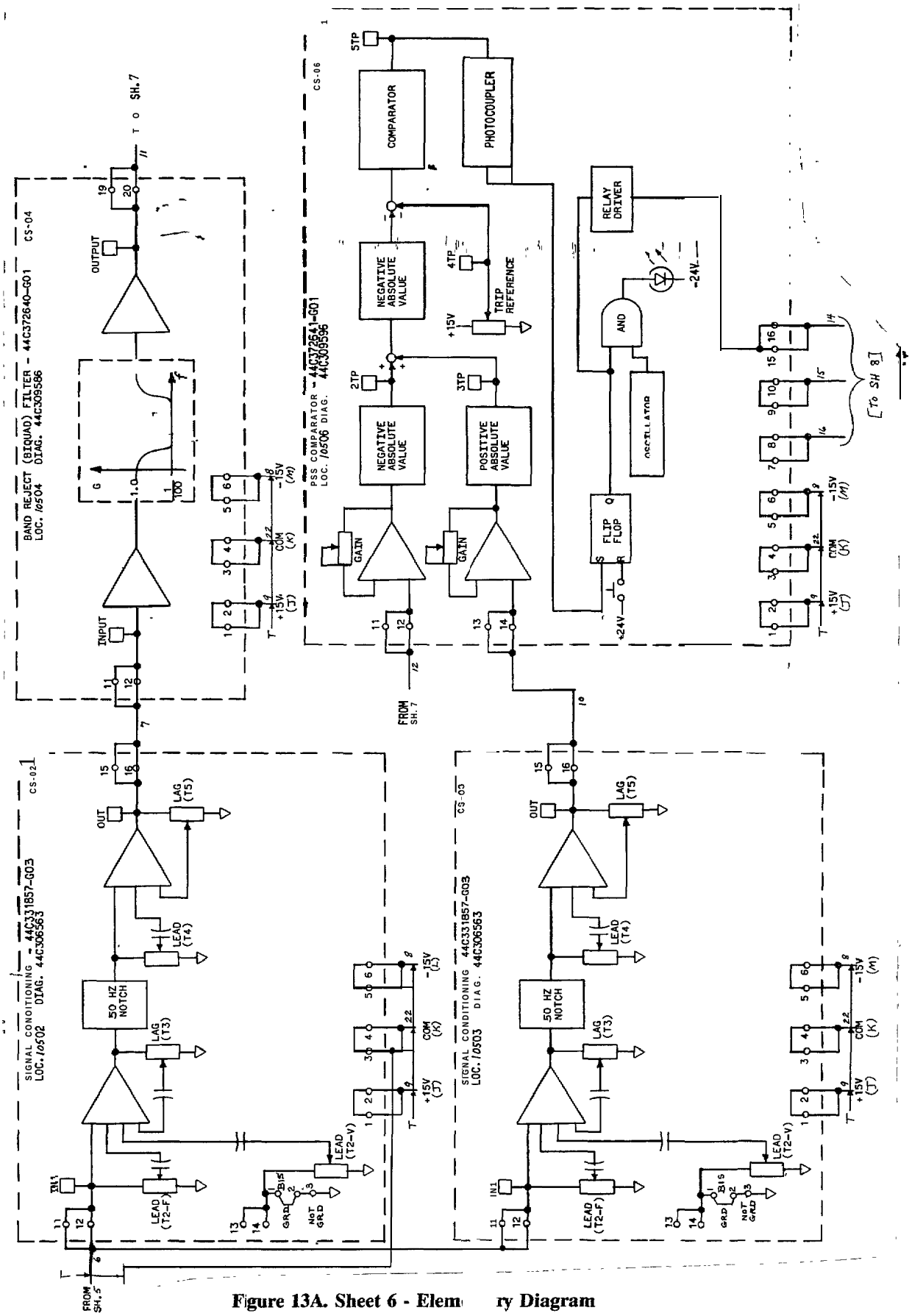


Figure 13A. Sheet 6 - Elem ry Diagram

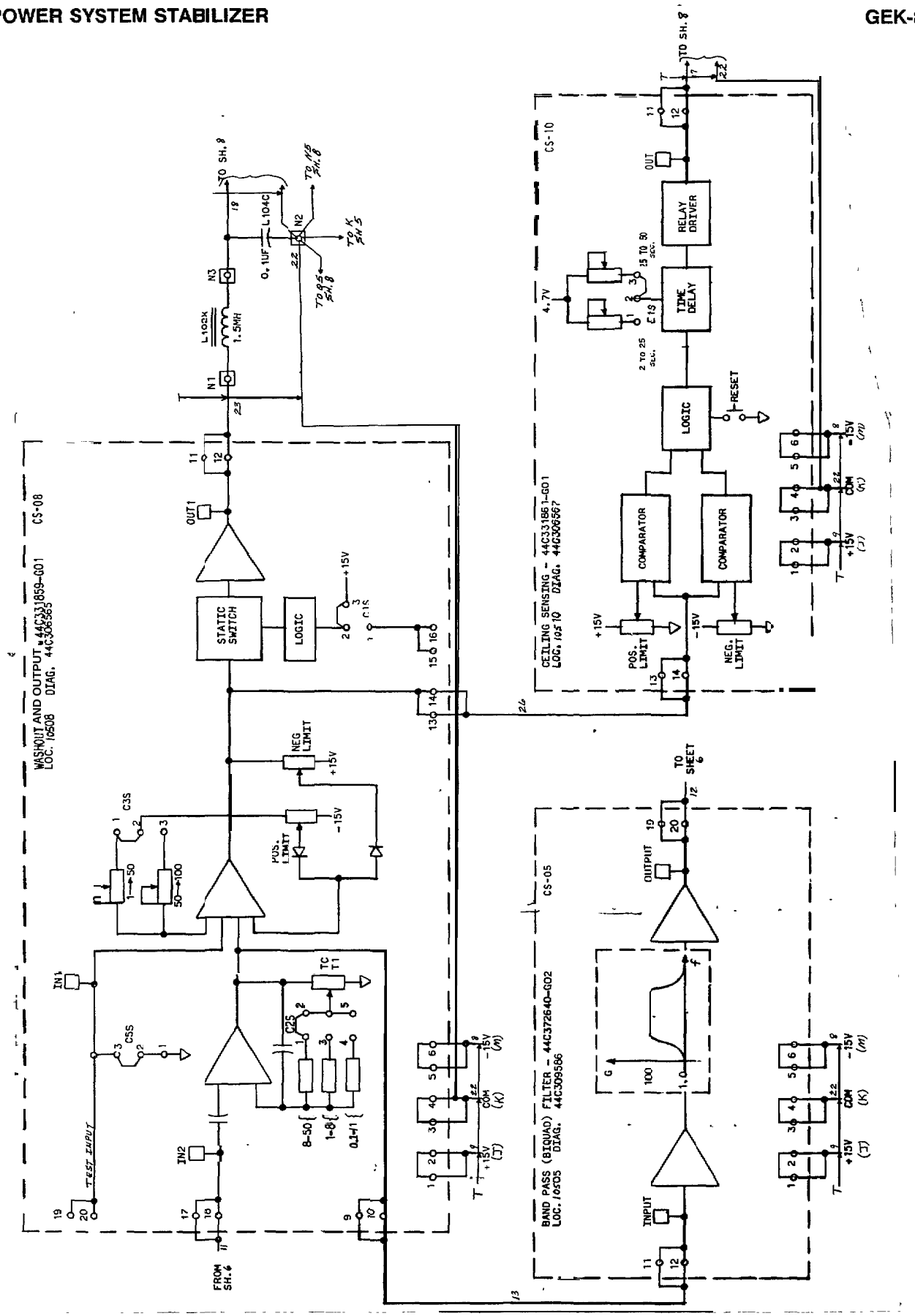


Figure 13B. Sheet 7 - Elementary Diagram

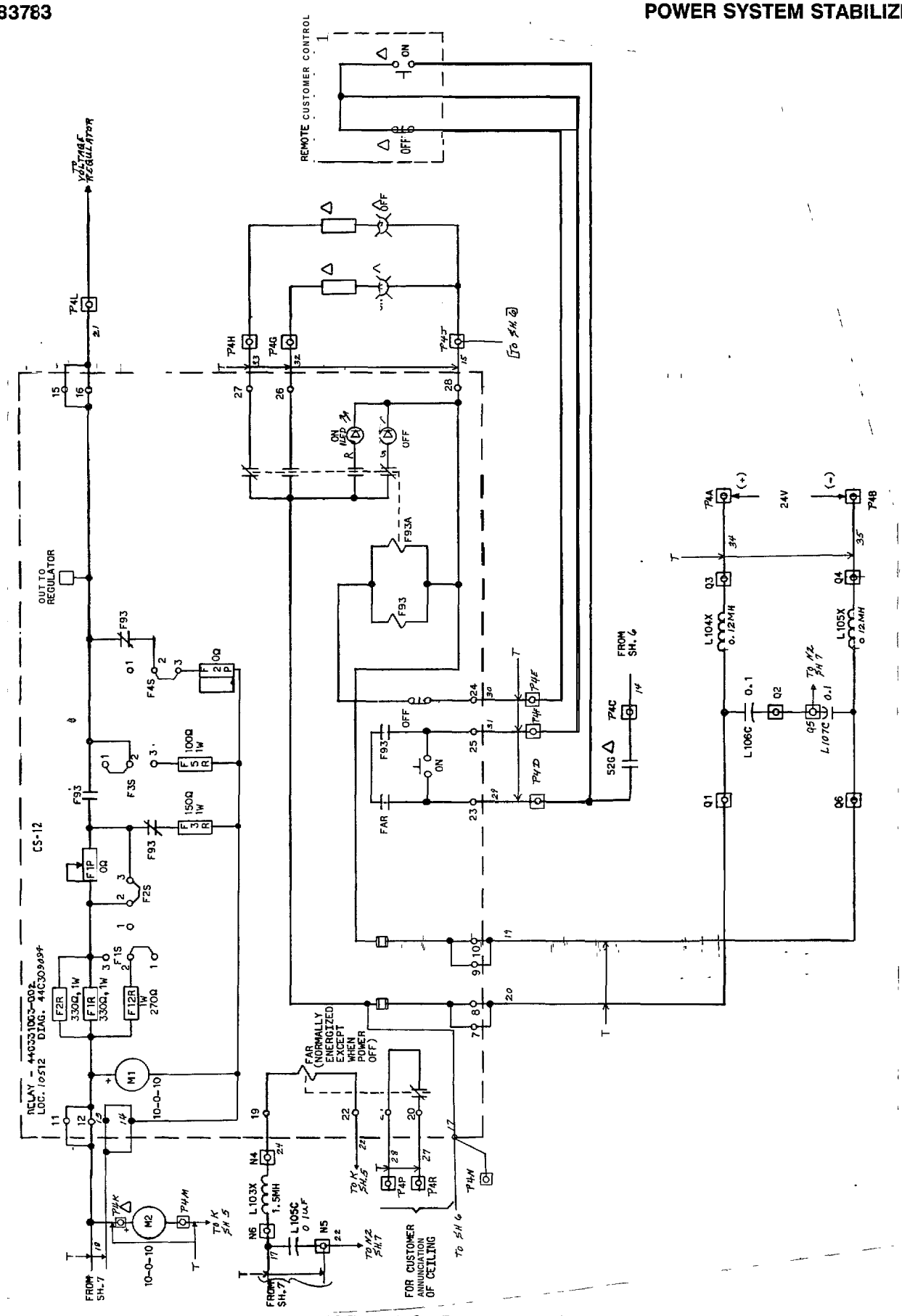


Figure 13C. Sheet 8 - Elementary Diagram

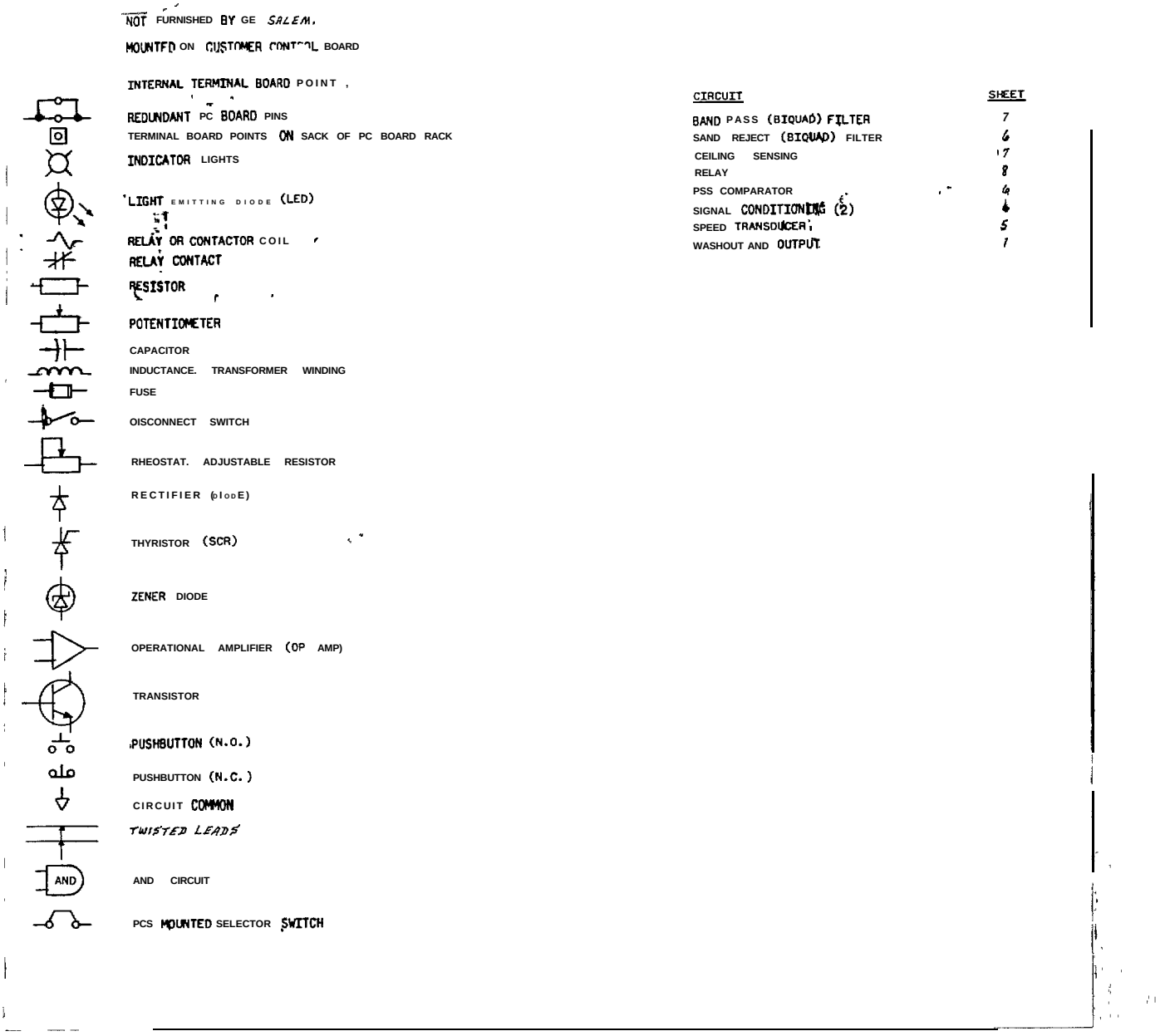


Figure 13D. Sheet 9 - Elementary Diagram

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